MATHEMATICS GRADE 10: AREA OF PART OF A CIRCLE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.3 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 2.0: Geometry
Sub-Strand	Sub-Strand 2.6: Area of a Part of a Circle
Total Duration	10 lessons × 40 minutes = 400 minutes total
Sub-Strand Content	– Arc length of a circle – Area of a sector of a circle – Area of a minor segment of a circle – Area of a major segment of a circle – Area of a major sector of a circle – Relationship between sector angle, radius, and area – Application of sector and segment areas in real-life contexts – Use of mathematical tables and calculators in computing circle part areas
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) identify and name parts of a circle including arc, sector, and segment in different situations, b) calculate the arc length of a circle given the radius and angle subtended at the centre, c) derive and apply the formula for the area of a sector of a circle using the central angle and radius, d) calculate the area of a minor sector and a major sector of a circle in different contexts, e) calculate the area of a minor segment and a major segment of a circle by subtracting the area of the relevant triangle from the sector area, f) apply knowledge of areas of parts of a circle to solve real-life problems such as land allocation, pizza slicing, and fan-shaped designs, g) appreciate the use of areas of parts of a circle in day-to-day activities and in various careers.
Core Competencies	– Communication and Collaboration: the learner discusses with peers to derive the formula for the area of a sector and shares strategies for solving problems involving parts of a circle, such as computing the area of a pizza slice or a fan-shaped maize field. – Critical Thinking and Problem Solving: the learner applies the sector and segment area formulas to solve multi-step real-life problems, including land allocation in the Rift Valley and designing curved rooftops in Nairobi. – Digital Literacy: the learner uses calculators and digital devices to compute areas of sectors and segments, and watches video demonstrations on how sector area formulas are derived and applied. – Creativity and Imagination: the learner designs fan-shaped or circular garden plots and calculates material requirements, connecting geometric reasoning to practical planning.

	– Self-Efficacy: the learner builds confidence in applying circle geometry formulas independently to unfamiliar real-world contexts.
Core Values	– Responsibility: the learner takes care of mathematical instruments (compasses, rulers, calculators) when constructing and measuring parts of a circle. – Unity: the learner collaborates in groups to derive formulas and solve problems involving areas of sectors and segments, appreciating diverse problem-solving approaches. – Respect: the learner listens to peers' explanations during group discussions on sector and segment area calculations and values different solution strategies. – Integrity: the learner honestly records measurements and calculations when determining the radius and angle of a pizza slice or a circular plot, and does not falsify results.
Science & Engineering Practices	– Asking Questions and Defining Problems: the learner poses questions about why different pizza slices of the same pie have different areas, driving inquiry into the relationship between central angle, radius, and sector area. – Using Mathematics and Computational Thinking: the learner applies the formula $(\theta/360°) \times \pi r^2$ to compute sector areas and uses subtraction of triangle area to find segment areas across multiple real-life scenarios. – Developing and Using Models: the learner draws and labels diagrams of circles showing sectors and segments, annotating radius, arc, and central angle to build a visual model that supports formula derivation. – Constructing Explanations: the learner uses evidence from measurements (radius of a pizza, angle of a slice) and calculations to explain why a larger central angle always produces a proportionally larger sector area. – Engaging in Argument from Evidence: the learner defends computed sector or segment areas using measured data and the formula, and critiques peers' calculations with reference to shared evidence.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Problem Solving): the learner applies sector and segment area calculations to make informed decisions, such as determining the fair share of a circular plot of land among family members in a rural Kenyan setting. – Life Skills (Creative Thinking): the learner designs circular or fan-shaped patterns for traditional Kenyan crafts and beadwork, computing the area of each coloured sector. – Financial Literacy: the learner estimates the cost of materials needed to tile or turf a circular garden in Kisumu by calculating the areas of its sectors and segments. – Environmental Education: the learner connects the efficient partitioning of circular irrigation fields (common in the Rift Valley) to water conservation and sustainable land use. – Safety: the learner handles geometric instruments such as compasses and protractors safely during construction and measurement activities.
Career Connections	– Architecture and Construction: architects designing curved roofs, circular windows, and domed structures in Nairobi rely on sector and segment area calculations to estimate material quantities. – Agriculture and Land Surveying: land surveyors in the Rift Valley use circular sector principles to partition fan-shaped irrigation plots and allocate land equitably. – Food Industry and Catering: caterers and bakers calculating the fair serving size of circular foods such as pizza, mandazi rings, or circular cakes apply sector area reasoning. – Engineering (Mechanical): engineers designing gear teeth, fan blades, and turbine components in Kenyan manufacturing plants use sector geometry in their calculations. – Graphic Design and Fashion: Kenyan fashion designers creating circular or fan-shaped fabric patterns and kitenge designs use sector area to minimise fabric waste. – Medicine (Radiology): medical imaging technicians computing cross-sectional areas of circular body structures apply sector and segment geometry concepts.

Focus for Lessons	This unit develops learners' ability to calculate the areas of parts of a circle — specifically sectors and segments — by building on their prior knowledge of circle properties, triangle area, and the constant π. Using the anchoring phenomenon of slicing a circular pizza (or mandazi), learners move from concrete measurement and observation to abstract formula derivation, ultimately applying their understanding to authentic Kenyan contexts such as land allocation, fan irrigation, and circular craft design. The unit integrates collaborative inquiry, digital tools, and model-building to ensure learners can confidently interpret and solve both straightforward and multi-step problems involving circular geometry.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we use the radius and the central angle of a circular pizza slice to calculate its exact area — and apply the same method to solve real problems in farming, design, and everyday life in Kenya? KICD KEY INQUIRY QUESTIONS: 1. How do we calculate the area of a part of a circle? 2. How do areas of sectors and segments of a circle apply in day-to-day activities?
Anchoring Phenomenon	A circular pizza (or a round chapati/mandazi) is placed on a table in front of the class and cut into unequal slices. Some slices are wide and some are narrow, yet all come from the same circular food of identical radius. Students immediately notice that the bigger-angled slices are larger in area, but are puzzled by exactly how much larger — is a slice with twice the angle always exactly twice the area? Is the relationship always perfectly proportional? Can we predict the exact area of any slice if we only know the radius of the pizza and the angle of the cut? This tangible, food-based scenario anchors all subsequent learning: learners measure the radius of the pizza base, use a protractor to measure the central angle of each slice, and then attempt to calculate its area — revealing the need for a precise formula. The phenomenon is deliberately puzzling because learners initially lack the formal tool (the sector area formula) to confirm or refute their predictions, motivating the entire unit's inquiry.
Storyline Thread	Lesson 1: ANCHORING PHENOMENON & PREDICT — Students observe unequal slices of a circular pizza/chapati; they predict which slice has the greatest area and estimate how area relates to the angle of the slice. The Driving Question Board is introduced and initial questions are posted. Lesson 2: OBSERVE & REVIEW — Students review parts of a circle (arc, sector, segment, chord) and measure the radius and central angle of physical circular cut-outs (representing pizza slices); they record data and look for a pattern between angle and area. Lesson 3: EXPLAIN — Arc Length — Students derive the formula for arc length ($\theta/360° \times 2\pi r$) through guided inquiry, linking the fraction of the full circle to the fraction of the circumference; Kenyan context: arc length of a centre-pivot irrigation sweep. Lesson 4: EXPLAIN — Area of a Sector (Minor) — Students derive the sector area formula ($\theta/360° \times \pi r^2$) by analogy with arc length; they apply it to calculate the area of a pizza slice, a maize shamba sector, and a bead panel, then verify with calculators. Lesson 5: EXPLAIN — Area of a Major Sector — Students extend the formula to major sectors (angle > 180°), distinguishing minor and major sectors; problem contexts include the larger unwatered portion of a circular irrigation field and the major fan of a matatu wiper. Lesson 6: OBSERVE & EXPLAIN — Area of a Minor Segment — Students observe that a segment is a sector minus a triangle; they subtract the area of the triangle ($\frac{1}{2}r^2\sin\theta$) from the sector area to find the minor segment area; context: the area of a watermelon rind slice. Lesson 7: EXPLAIN — Area of a Major Segment — Students calculate the area of a major segment as the total circle area minus the minor segment area; they solve problems involving land allocation of circular plots in the Rift Valley. Lesson 8: APPLICATION & MODEL BUILDING — Students apply all formulas (arc length, sector area, segment area) to multi-step real-life problems including tiling a circular garden, pricing watermelon slices fairly, and designing a Maasai bead pattern; models are revised on the DQB. Lesson 9: DIGITAL TOOLS & CONSOLIDATION — Students use calculators and digital simulations to confirm earlier hand-calculated sector and segment areas; they identify and correct common errors (e.g., using degrees vs. radians, forgetting to

	subtract the triangle); DQB is updated. Lesson 10: ASSESSMENT & REFLECTION — Students complete a summative task: given the radius and angle of a sector of a circular shamba, they calculate the arc length, sector area, and segment area, then present their solutions and reflect on how the anchoring phenomenon (pizza slicing) connects to all learned concepts; the DQB is finalised.
Supporting Phenomena	– A circular maize shamba (farm) in the Rift Valley irrigated by a centre-pivot sprinkler system: the sprinkler sweeps a sector-shaped area of crops — farmers need to know the exact area watered to plan fertiliser and seed quantities. – A Maasai bead necklace featuring repeating fan-shaped (sector) colour panels: artisans must calculate the area of each coloured sector to purchase the correct quantity of beads of each colour. – A slice of a round watermelon (tikiti maji) sold by vendors along Mombasa Road: vendors price their slices by estimated area, raising the question of whether they are charging fairly relative to the size of the central angle cut. – The swept area of a windshield wiper on a matatu: the wiper clears a ring-sector (annular sector) shape — mechanics and engineers need sector area to determine the effective cleaning zone and select correct wiper blade sizes.

LESSON 1 (80 minutes): Anchoring Phenomenon & Predict — Unequal Slices, Unequal Areas	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon — a circular pizza/chapati cut into unequal slices — and to elicit students' prior knowledge and predictions about how the area of a slice relates to its central angle and radius, motivating the need for a precise sector-area formula.
Knowledge	Students will recall the formula for the area of a full circle ($A = \pi r^2$) and identify that a slice of a circle (sector) is a fraction of the whole circle determined by its central angle.
Skills	Students will measure the radius of a circular object and the central angle of a sector using a ruler and protractor, record observations systematically, and construct proportional estimates for sector area.
Attitudes	Students will appreciate the relevance of circle geometry in everyday Kenyan life — from sharing food equitably to land apportionment in farming — and show curiosity and intellectual honesty when their predictions are uncertain or contradicted.
Key Inquiry Question	How do we calculate the area of a part of a circle?
Purpose in Storyline	This is the opening lesson of the unit. It anchors all subsequent learning in a single tangible puzzle: a circular pizza/chapati cut into unequal slices. By predicting, measuring, and debating before any formula is taught, students experience the cognitive need for the sector-area formula — setting up every lesson that follows.
Safety Notes	If a real pizza or chapati is used, ensure the cutting knife is handled only by the teacher. Students should not handle sharp implements. Ensure no student has a food allergy to wheat/gluten before using chapati as the demonstration object. If allergies are a concern, use a cardboard circle instead.

B. LESSON OVERVIEW

The teacher places a large circular pizza base (or a round chapati) on a central demonstration table and cuts it — in full view of the class — into five unequal slices using pre-marked central angles (approximately 30°, 45°, 60°, 90°, and 135°). Students immediately notice the visible size differences and are prompted to predict: which slice has the greatest area, and is a slice with twice the angle always exactly twice the area? Working in groups of four, learners measure the radius of the pizza and the central angle of each slice with a ruler and protractor, then use their knowledge of the full-circle area formula ($A = \pi r^2$) to make proportional estimates. Groups record predictions on sticky notes and post their biggest unresolved questions on the newly opened Driving Question Board (DQB). No formula is formally taught in this lesson — the lesson closes with productive uncertainty and a shared list of questions the class will investigate across the unit.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Fractions on a number line] Source: Khan	Students sit in groups of four. The teacher places a large circular pizza base (or round chapati, radius ≈ 15 cm) on the demonstration table. Before any	Before cutting, say: 'Look carefully at this chapati. It is perfectly circular — just like the round flat breads many of our mothers and grandmothers make at home in	Strategy: PREDICT-BEFORE-OBSERVE (Elicitation of Prior Knowledge). By committing to a written prediction before any instruction, students activate their	OBSERVABLE INDICATOR: Collect and scan the group prediction sheets. Look for two things — (1) whether students correctly rank the slices by angle

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Proportions with Angle Bisectors (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	cutting, each student silently sketches the circle in their exercise book and writes a personal prediction: 'If I cut this circle into slices with angles 30°, 45°, 60°, 90°, and 135°, rank the slices from smallest to largest area. Do you think a 90° slice is exactly three times the area of a 30° slice? Why or why not?' After 4 minutes of silent individual thinking, students share predictions within their group and agree on a group prediction statement. Each group writes their prediction on a half-sheet of paper. Groups then do a gallery walk (3 minutes) to read other groups' predictions before returning to their seats. Learners then watch as the teacher cuts the pizza/chapati into the five pre-marked slices and the slices are laid out on the table, clearly labelled by angle.	Nairobi, Kisumu, or Mombasa. I am about to cut it into five slices — but the slices will NOT be equal. Your job right now is to predict, before we do any mathematics.' Distribute the silent prediction prompt on the board or printed sheet. Allow 4 minutes of WAIT TIME — do not accept shouted answers. After group discussion, cold-call EXACTLY 3 groups to share their group prediction: 'Group 2 — tell us your prediction and your reasoning.' After the gallery walk, say: 'Notice that not all groups agree. That disagreement is exactly why we need mathematics. Let us now cut the chapati and see what we are actually dealing with.' Cut the pizza/chapati live, using a protractor and marker to pre-mark the angles. Lay the five slices on labelled paper plates or sheets of paper showing their angles.	existing mental models of area and proportion. The disagreement between groups creates cognitive dissonance — a necessary precondition for learning. This strategy surfaces the common misconception that area and angle are always proportional, which will be confirmed (it is true for sectors) but needs formal proof, motivating the formula.	(demonstrating understanding that larger angle → larger area), and (2) whether students claim the relationship is exactly proportional (e.g., '90° slice = 3× the 30° slice'). Students who cannot rank the slices need immediate support recalling what a central angle is. Students who confidently assert exact proportionality without justification are ready to be challenged with the formal verification in Phase 2.
Observe Phase 📹 **VIDEO:** Fractions on a number line Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Proportions with Angle Bisectors (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group receives: one of the five labelled pizza/chapati slices (or a cardboard sector cut to the same dimensions), a ruler, a protractor, a calculator, and a recording table with columns: Slice Label | Central Angle (°) | Measured Radius (cm) | Fraction of Full Circle (θ/360°) | Estimated Area (cm²) using $A = (\theta/360°) \times \pi r^2$. Students measure the radius (from the point/tip of the slice to the curved edge) and verify the central angle with their protractor. They then compute the fraction of the full circle their slice represents and multiply by the full-circle area (πr^2) to get an estimated sector area. Groups record all working clearly. The teacher circulates and takes photos of student recording tables for later class discussion.	Distribute materials and say: 'Your task is to measure your slice carefully and use what you already know about circles to ESTIMATE its area. You have one tool available: the area of a full circle is $A = \pi r^2$. Think about what fraction of the full circle your slice is.' Allow 12 minutes of group work. Circulate and ask probing questions to each group — do not give answers: 'What fraction of the full 360° does your slice represent?', 'How does that fraction help you think about what fraction of the total area your slice should be?', 'What would the area of the WHOLE chapati be if its radius is 15 cm?' After 12 minutes, cold-call EXACTLY 4 groups to explain their calculation method aloud (not just the answer). Say:	Strategy: DATA COLLECTION & PATTERN RECOGNITION. By physically measuring their own slice and computing its area using proportional reasoning — before the formal formula is named — students construct the sector area formula ($A = (\theta/360°) \times \pi r^2$) inductively from first principles. Seeing all five calculated areas displayed simultaneously on the demonstration table allows students to visually confirm the proportional relationship and check whether their prediction was correct. This 'discovery before definition' approach builds durable conceptual understanding.	OBSERVABLE INDICATOR: Circulate and check whether students correctly identify the radius as the distance from the tip (centre of the original circle) to the curved edge — NOT the arc length. A common error is measuring the arc length instead of the radius. Also check that students compute $(\theta/360°) \times \pi r^2$ and not $(\theta/360°) \times 2\pi r$ (confusing area with circumference). Groups that make this error need immediate targeted questioning: 'Are we looking for a length or a surface? What is the unit of area?'

	After calculations, each group writes their sector area on a large sticky note and places it next to their slice on the demonstration table so the whole class can see all five values simultaneously.	'Group 3 — walk us through your reasoning step by step.' After all five sticky notes are posted, ask the whole class: 'Does anyone notice anything interesting about the relationship between the angles and the areas? Is a 90° slice exactly three times a 30° slice? Check with your calculator now — 10 seconds of WAIT TIME.'		
Explain Phase 📹 **VIDEO:** [Fractions on a number line] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Proportions with Angle Bisectors (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class discussion anchored by the five area values now displayed on the demonstration table. Students are guided to articulate the pattern: the area of a sector is always the same fraction of the full circle area as the central angle is of 360°. Students write the generalised formula in their own words first, then in symbolic form: $A = (\theta/360°) \times \pi r^2$. Each student individually applies the formula to one new problem in their exercise book: 'A circular shamba (farm plot) in the Rift Valley has a radius of 20 metres. A farmer fences off a sector-shaped portion for planting maize, with a central angle of 72°. What is the area of the maize plot? Use $\pi = 3.142$.' Students self-check by comparing with a partner.	Point to the five sticky notes and say: 'Look at what your measurements and calculations have shown. Let us read the pattern together.' Cold-call 2 students to state the pattern in words. Then say: 'Mathematics gives this pattern a name. We call this slice of a circle a SECTOR. And the rule you have just discovered — that its area is the fraction (θ/360°) of the full circle — is the Sector Area Formula.' Write on the board: $A = (\theta/360°) \times \pi r^2$. Say: 'This formula works for any sector — a pizza slice in Nairobi, a farm plot in the Rift Valley, or a design element on a Maasai shield. Anywhere there is a circular sector, this formula gives its exact area.' Issue the individual shamba problem. Allow 5 minutes of silent individual working. After 5 minutes, cold-call 1 student to put their working on the board. Say: 'Before we move on — turn to your partner and explain in one sentence WHY we divide by 360.' Allow 15 seconds of WAIT TIME before cold-calling.	Strategy: CONNECT-EXTEND-CHALLENGE (Visible Thinking Routine). Students first CONNECT the formula to what they already computed in Phase 2 (validating their prior reasoning), then EXTEND it to the new Rift Valley shamba context (transfer to a Kenyan real-world scenario), then are CHALLENGED to explain why 360° appears in the denominator (deepening conceptual understanding beyond procedural memorisation). Naming the formula only AFTER students have derived it inductively ensures the symbol is attached to meaning.	OBSERVABLE INDICATOR: Review the individual shamba calculations. Students who arrive at $A = (72/360) \times 3.142 \times 20^2 = 251.36\ m^2$ (approximately) are on track. Watch for two errors: (1) students who use diameter instead of radius ($r = 20$ is stated, so this should not occur, but check), and (2) students who forget to square the radius. Any student who cannot set up the formula correctly needs a re-teaching conversation before the next lesson.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Fractions on a	The teacher formally introduces the Driving Question Board — a dedicated section of the classroom display wall (or a large sheet of chart paper) headed with the unit's driving question: 'How can we use	Say: 'This board will be our class investigation wall for the entire unit. Every question you post today is a real mathematical question — there are no wrong questions. Our job over the next	Strategy: DRIVING QUESTION BOARD (DQB) — Metacognitive Anchoring. The DQB externalises student thinking and makes the inquiry public and collective. By distinguishing 'what I now	OBSERVABLE INDICATOR: Read all orange sticky notes after class. Categorise them into: (a) questions that reveal correct understanding but curiosity about extension (e.g., 'Does this work for

[number line](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Proportions with Angle Bisectors (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	the radius and the central angle of a circular pizza slice to calculate its exact area — and apply the same method to solve real problems in farming, design, and everyday life in Kenya?' Each student receives two sticky notes. On the FIRST sticky note (blue): they write one thing they now understand about area and sectors. On the SECOND sticky note (orange): they write one question they still have — something the pizza/chapati phenomenon made them wonder about that they cannot yet answer. Students post both sticky notes on the DQB. The teacher reads out selected questions aloud, clusters them into emerging themes (e.g., 'What about irregular shapes?', 'What if the slice is not a perfect triangle — how do we find the leftover area?'), and labels the clusters. These clusters will be revisited and resolved in subsequent lessons.	lessons is to answer every single one of them using the mathematics of circles.' Allow 3 minutes for students to write both sticky notes silently. As students post notes, circulate and read them. Select 3–4 orange (question) sticky notes that preview upcoming content (e.g., questions about segments, about non-pizza real-life contexts, about what happens when θ = 180°) and read them aloud: 'Listen to this question from a classmate: [read question]. Who else is wondering the same thing?' Show of hands. Say: 'We will answer this in Lesson 3. Let us hold on to that curiosity.' Explicitly cluster the questions into 2–3 labelled groups on the DQB using a marker directly on the chart paper.	understand' (blue) from 'what I still wonder' (orange), students practise metacognitive monitoring — a core competency in CBE. The teacher's act of clustering questions signals to students that their wondering has mathematical structure, building academic self-esteem and investment in the unit.	semicircles?'), (b) questions that reveal persistent misconceptions (e.g., 'Is the area the same as the perimeter of the slice?'), and (c) questions that preview the next concept accurately (e.g., 'What is the area of just the triangular part of the slice without the curved bit?'). Use category (b) questions to plan targeted re-teaching at the opening of Lesson 2.
Model Building Phase 📹 **VIDEO:** [Fractions on a number line](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Proportions with Angle Bisectors (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	Students individually draw an INITIAL MODEL in their exercise books. The prompt is: 'Draw a diagram that shows a circular pizza of radius r cut into a sector with central angle θ. Label all the parts you know. Write the formula for its area. Show using arrows or annotations WHY the formula makes sense — connect each part of the formula to a part of your diagram.' Students annotate their drawings with labels: centre of circle, radius r, central angle θ (in degrees), arc, sector. They write $A = (\theta/360°) \times \pi r^2$ next to the sector and add an annotation explaining what each part of the formula means (e.g., 'θ/360° = the fraction of the full circle that this slice	Say: 'In this unit, you will keep an evolving diagram — a scientific and mathematical model — that grows with your understanding. Today's model is your STARTING POINT. It does not need to be perfect; it needs to be honest — show exactly what you understand right now, no more, no less.' Allow 7 minutes of silent individual drawing and annotating. Circulate and look for models that show the full circle alongside the sector (not just the sector in isolation) — this signals the student understands the sector as a fraction of the whole. After 7 minutes, cold-call 1 student to share their model under a document camera or by holding it up. Say: 'Notice how [student	Strategy: ANNOTATED DIAGRAM / INITIAL MODEL. Having students build a labelled visual model at the end of the first lesson serves two purposes: it consolidates today's learning into a single coherent image, and it creates a baseline artefact that will be progressively revised — making learning growth visible. The annotation requirement ('explain WHY the formula makes sense') prevents students from copying a formula without meaning, a critical safeguard for conceptual understanding.	OBSERVABLE INDICATOR: Collect or photograph initial models. A proficient model will show: a full circle with a clearly marked centre, a sector shaded and labelled with θ and r, the formula $A = (\theta/360°) \times \pi r^2$, and at least one annotation connecting the formula to the diagram. A developing model shows the sector but lacks the annotation or the connection to the full circle. A beginning model shows only the formula with no diagram. Use this three-tier classification to form targeted support groups for Lesson 2.

	represents'). These initial models are dated and will be revised in every subsequent lesson as understanding deepens.	name] has drawn the full circle AND the sector together. That connection — sector as part of a whole circle — is the most important idea you will carry into every lesson this unit.' Close the lesson by returning to the phenomenon: 'Next lesson, we will go deeper. Today you predicted and measured. Soon you will be able to calculate the exact area of any sector — any slice of any circle — anywhere in Kenya.'		

D. TEACHER REFLECTION

After the lesson, consider the following reflective questions: (1) Did students' predictions reveal the proportionality misconception (e.g., assuming area and angle are always linearly related)? If so, was it sufficiently addressed through the Phase 2 calculations, or does it need revisiting in Lesson 2? (2) Were students able to correctly identify the radius of the sector (from the tip/centre to the curved edge) as opposed to measuring the arc length? If more than one-third of groups made this error, plan a 5-minute physical demonstration at the start of Lesson 2 using a string to trace the difference between radius and arc. (3) Review the orange DQB sticky notes: are there questions about the 'leftover' part of the sector (the triangular region vs. the circular segment)? If yes, the storyline is progressing as intended — students are already anticipating the concept of a circular segment. (4) Were the Kenyan contexts (chapati, Rift Valley shamba) effective at sustaining engagement? Note any suggestions students made for alternative local contexts (e.g., a round borehole cover, a fan-shaped maize field boundary) and consider incorporating them in Lessons 2–4. (5) Did the DQB launch feel natural and purposeful, or did it feel procedural? If students seemed unsure what to write on their sticky notes, in the next lesson begin by modelling a sample question: 'Here is the kind of question a curious mathematician asks after today: What if the angle is 180° — is the area exactly half the circle? How would we check?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A circular pizza/chapati was cut into five unequal slices with central angles of approximately 30°, 45°, 60°, 90°, and 135°. The slices were visibly different in size, with the wider-angled slices clearly larger. All slices shared the same radius.
What did I learn?	The area of a sector (a slice of a circle) is a fraction of the full circle's area, determined by the ratio of its central angle to 360°. The formula is $A = (\theta/360°) \times \pi r^2$, where θ is the central angle in degrees and r is the radius. A sector with twice the central angle has exactly twice the area — the relationship is directly proportional.
How does this explain the phenomenon?	The sector area formula works because a full circle corresponds to a central angle of 360°. Any sector with central angle θ represents the fraction $\theta/360°$ of the full circle. Multiplying this fraction by the full circle area (πr^2) gives the exact area of the sector. This same principle applies to real contexts such as a fan-shaped maize plot in the Rift Valley, a slice of round chapati, or any circular design element.

LESSON 2 (80 minutes): OBSERVE & REVIEW — Parts of a Circle, Radius, Central Angle, and the Hunt for a Pattern	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to identify and name parts of a circle (arc, sector, segment, chord, radius, diameter), measure the radius and central angle of circular cut-outs representing pizza/chapati slices, record and organise data in a table, and begin to detect the proportional pattern between central angle and area — building the evidence base needed to derive the sector area formula in later lessons.
Knowledge	Learners will know: (a) the names and definitions of the parts of a circle — arc, sector, segment, chord, radius, diameter, and central angle; (b) that the full angle at the centre of a circle is 360°; (c) that the area of a full circle is πr^2; (d) that a sector is a 'pizza-slice' shaped region bounded by two radii and an arc; (e) that doubling the central angle doubles the sector area (proportionality), revealed through measured data.
Skills	Learners will be able to: (a) correctly label a diagram of a circle with all required parts; (b) measure the radius of a circular cut-out using a ruler; (c) measure the central angle of a sector cut-out using a protractor; (d) compute the area of the full circle using πr^2; (e) compute the fractional part of the circle represented by each sector ($\theta/360°$); (f) record data neatly in a structured table; (g) identify and articulate the proportional pattern from collected data.
Attitudes	Learners will appreciate: (a) the value of careful, precise measurement in arriving at reliable mathematical conclusions; (b) how everyday Kenyan food objects (chapati, pizza, mandazi) carry embedded mathematical structures; (c) the importance of honest, accurate data recording as the foundation of mathematical inquiry.
Key Inquiry Question	How do we calculate the area of a part of a circle?
Purpose in Storyline	In Lesson 1 (Predict Phase), learners predicted whether a wider pizza slice is always proportionally larger and recorded intuitive guesses. In this Lesson 2 (Observe & Review Phase), learners move from guessing to gathering real evidence: they handle physical circular cut-outs (sector shapes representing pizza/chapati slices), review the formal vocabulary of circle parts, take measurements, and fill a data table that will be the launchpad for deriving the area formula in Lesson 3. This lesson answers the first layer of the driving question: YES — the relationship IS proportional — and sets up the deeper question of HOW to express that proportionality as a precise formula.
Safety Notes	Scissors are used to prepare cut-outs before the lesson (teacher-prepared). If learners handle scissors during the activity, remind them to carry scissors with the blade pointing downward and to cut away from the body. Compasses used for drawing circles must be handled with the point directed downward onto the paper at all times. No food is to be consumed during the mathematics lesson; the chapati/pizza is a display model only.

B. LESSON OVERVIEW
Learners begin by revisiting their Lesson 1 predictions about pizza/chapati slices posted on the Driving Question Board. The teacher then leads a rapid, vocabulary-building review of circle parts using a large labelled diagram. The heart of the lesson is a hands-on DATA COLLECTION ACTIVITY: groups receive sets of pre-cut circular sector shapes (cut from A3 card circles of the same radius, representing different pizza slices) and measure each sector's radius and central angle with rulers and protractors. They compute the full-circle area and the fractional part each sector represents, record results in a structured table, and compare across sectors. Through guided discussion, learners surface the proportional pattern (a sector with twice the angle has twice the area). The lesson closes with learners updating the Driving Question Board with new evidence and revising their cumulative circle model to include labelled parts and the fraction concept ($\theta/360°$).

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Proof: radius is perpendicular to a chord it bisects Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Proportions with Angle Bisectors (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners re-read their Lesson 1 predictions from the Driving Question Board about whether a slice with twice the angle is always exactly twice the area of the smaller slice. Each learner silently re-examines their written prediction and marks it with a ✓ (still agree), ? (unsure), or ✗ (now disagree) based on any new thinking since Lesson 1. Learners then share their self-assessments with a shoulder partner for 2 minutes. Two or three learners share their reasoning aloud with the class. This reconnects the lesson to the anchoring phenomenon (the unequal chapati/pizza slices) and creates cognitive readiness to gather real evidence.	1. Display the physical circular chapati/pizza model from Lesson 1 at the front of the class. Point to two unequal slices and say: 'Last lesson you made a prediction. Today we gather EVIDENCE. Let's see if your prediction was right.' 2. Direct learners to their Lesson 1 prediction cards/notebooks: 'Take 30 seconds — silently re-read your prediction and mark it ✓, ?, or ✗.' [WAIT 30 seconds — do not speak.] 3. Say: 'Turn to your shoulder partner. Tell them: what did you predict, and has your thinking shifted at all? You have 2 minutes.' [Circulate, listen for the proportionality idea.] 4. Cold-call exactly 3 learners (one ✓, one ?, one ✗): 'Amina, what did you predict and what is your mark now?' [WAIT 10 seconds after each name before prompting.] 5. Record the three positions on the board as a tally. Say: 'By the end of today's lesson, your evidence will tell you who was right. Let's get to work.'	PREDICTION REVISIT ('Commit & Reconsider'): Learners publicly re-commit to or revise their prior prediction before gathering data. This strategy activates prior knowledge, surfaces existing misconceptions (e.g., 'the area depends on the shape of the slice, not the angle'), and creates a personal stake in the evidence-gathering activity that follows. It also models scientific and mathematical practice: claims must be tested against evidence.	OBSERVABLE INDICATOR: When cold-called, can the learner articulate a reason for their ✓/✗/? mark — even an informal one (e.g., 'I think twice the angle should give twice the area because it is twice as much of the circle')? Teacher listens for whether learners are reasoning proportionally or are still guessing arbitrarily. Learners who say 'I just don't know' are flagged for targeted support during the data collection activity.
Observe Phase 📹 **VIDEO:** Proof: radius is perpendicular to a chord it bisects Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Proportions with	The teacher displays a large, unlabelled diagram of a circle with a sector and a segment drawn on it. Learners work in pairs to label as many parts as they can recall from their prior learning (radius, diameter, chord, arc, sector, segment, central angle). After 3 minutes, pairs compare labels with the pair behind them (groups of 4). The teacher then leads a whole-class reveal, one part at a time, using a colour-coded diagram.	1. Project or draw a large unlabelled circle diagram. Say: 'You have 3 minutes — working with your partner, label every part of this circle that you remember. Use pencil.' [WAIT 3 minutes. Circulate. Note pairs who cannot recall 'segment' vs 'sector' — these are the critical misconceptions to address.] 2. Say: 'Now compare your labels with the pair directly behind you. Where do you agree? Where do	LABEL-COMPARE-CORRECT ('Vocabulary Excavation'): Learners first attempt labelling from memory, then negotiate with peers, then receive expert confirmation. This three-stage sequence means new vocabulary is anchored to prior knowledge rather than introduced cold. Writing personal one-sentence definitions forces learners to process meaning rather than copy words — a critical move for	OBSERVABLE INDICATOR: In the '8 equal slices → 45°' mini-whiteboard check, teacher scans for: (a) correct answer 45° [target], (b) common error 360/8 = 40 [arithmetic slip], (c) answer of 90° [learner divided by 4 instead of 8]. Any learner showing (c) needs immediate peer correction. Teacher notes which pairs could not distinguish sector from segment during the labelling — these pairs are seated near a

Angle Bisectors (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners correct and complete their own diagrams in their exercise books. Special emphasis is placed on SECTOR (the 'pizza-slice' region bounded by two radii and an arc) and CENTRAL ANGLE (the angle at the centre between the two radii) — the two quantities they will measure in the next phase. Learners write a one-sentence definition of each part in their own words.	you disagree? 2 minutes.' [WAIT 2 minutes.] 3. Reveal labels ONE AT A TIME on the board. For each part, cold-call a learner to read their definition: 'Kipchoge, read me your one-sentence definition of a SECTOR.' [WAIT 10 seconds.] 4. After revealing SECTOR and CENTRAL ANGLE, hold up one of the pre-cut sector card pieces and say: 'This card shape — IS this a sector? How do you know?' [WAIT 10 seconds. Cold-call 2 learners.] 5. Critical bridging question — ask the whole class: 'If I cut a chapati into 8 equal slices, what is the central angle of each slice? Show me on your mini-whiteboard / slate / piece of paper.' [WAIT 15 seconds. Scan responses. Correct answer: 360 ÷ 8 = 45°.] 6. Explicitly connect to phenomenon: 'Every pizza or chapati slice you have ever eaten is a SECTOR. The central angle is the angle of the cut. That is exactly what we are about to measure.'	English Language Learners and learners in rural schools who may not have seen formal circle diagrams recently.	stronger group during Phase 3.
Observe Phase 📹 **VIDEO:** Proof: radius is perpendicular to a chord it bisects Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Proportions with Angle Bisectors (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of 4 receive an ACTIVITY ENVELOPE containing: (a) 4 pre-cut sector-shaped cards (all cut from circles of the same radius r = 10 cm, but with different central angles: 45°, 90°, 135°, 180°), (b) a ruler, (c) a protractor, (d) a DATA COLLECTION TABLE printed on A4 paper. Learners measure the radius and central angle of each sector card independently, then cross-check within the group. They record data in the table: Sector Label | Measured Radius (cm) | Measured Central Angle (°) | Full Circle Area = πr² | Fraction of Circle = θ/360° | Estimated Sector Area = (θ/360°) × πr². Learners	1. Distribute envelopes. Say: 'Before you open the envelope — your job is to MEASURE, RECORD, and SPOT THE PATTERN. You are detectives. The chapati slices are your evidence. You have 20 minutes.' 2. [At 2 minutes] Circulate to check measurement technique. Say to any group holding the protractor incorrectly: 'Where is the centre point of your protractor? It must sit exactly on the corner of the sector — that corner IS the centre of the original circle.' 3. [At 5 minutes] Cold-call one group: 'What radius did you measure for Sector A? Is that the	DATA TABLE + GRAPHING ('Pattern Detection through Systematic Recording'): Learners move from raw measurements to computed values to a visual graph. This three-layer representation (numerical table → computed fraction → graph) is a core NGSS Practice 4 (Analyzing and Interpreting Data) move. The graph makes the LINEAR proportionality unmistakably visible — the straight line through the origin is the most powerful piece of evidence they will carry into the Explain phase. The structured table also scaffolds algebraic thinking: learners see the formula	OBSERVABLE INDICATORS: (1) Measurement accuracy — sector cards are pre-labelled with correct answers on the back; groups self-check after completing the table. A group with radius error > 0.5 cm or angle error > 3° is prompted to re-measure. (2) Proportionality sentence — teacher collects the written sentence from each group at the end of Phase 3. Acceptable evidence of understanding: 'Yes, the 90° sector has exactly twice the area of the 45° sector because 90/360 is twice 45/360.' Incomplete understanding: 'The bigger angle gives more area' (true but not quantitatively precise).

	use π ≈ 3.14 and compute each column step-by-step. After completing the table, each group plots a simple graph: Central Angle (x-axis, 0–360°) vs Sector Area (y-axis). They observe whether the relationship is linear (a straight line through the origin). The group discusses and answers: 'Is a slice with twice the angle always exactly twice the area?'	same for Sector B?' [Expected: yes, all 10 cm — confirming controlled variable.] Say: 'Why does it matter that all sectors have the same radius? Tell the person next to you.' [WAIT 10 seconds.] 4. [At 12 minutes] Pause the class for 60 seconds. Say: 'Hands up — who has spotted something interesting in their table already? Do NOT tell me the answer yet — just say YES or NO.' [Count hands aloud.] Say: 'Hold that thought. Finish the table first.' 5. [At 18 minutes] Direct groups to the graphing task: 'Plot your four points. Draw the best straight line you can through them. Does it pass through the origin? What does that mean?' [WAIT 15 seconds. Circulate.] 6. [At 22 minutes] Say: 'I want every group to write one sentence at the bottom of their table answering: Is a slice with twice the angle always exactly twice the area? How do you know?' [WAIT 3 minutes for writing.]	$A = (\theta/360°) \times \pi r^2$ emerging naturally from the columns they fill in.	Misconception: 'No, it depends on the shape' — these groups need re-teaching in Phase 4.
Explain Phase 📹 **VIDEO:** Proof: radius is perpendicular to a chord it bisects Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Proportions with Angle Bisectors (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups share their data tables and graphing results in a whole-class gallery-walk style: each group's table is posted on the wall and all learners do a 5-minute silent walk to read other groups' tables, placing a sticky dot next to any result that surprises them. The class then reconvenes for a teacher-led discussion that draws out the proportionality principle explicitly. Learners are guided to see that the sector area formula $A = (\theta/360°) \times \pi r^2$ is not a new, mysterious rule — it is exactly what their data table computed, column by column. Learners copy the formula into their books,	1. Post all group tables on the wall. Say: 'You have 5 minutes to walk silently and read every group's table. Place a sticky dot on any number that surprises you.' [WAIT 5 minutes. Observe where dots cluster.] 2. Reconvene. Point to the most-dotted result. Say: 'Why did so many of you dot this number? What surprised you?' [Cold-call 2 learners. WAIT 10 seconds each.] 3. Draw a clean table on the board with the class's data. Ask: 'Look at Column 5 — Fraction of Circle. What operation did we do? Theta divided by 360. Now look at Column 6 — Sector Area.	GALLERY WALK + FORMULA CONSTRUCTION ('Evidence to Abstraction'): The gallery walk externalises all groups' data simultaneously, allowing learners to see that every group's table points to the same pattern — building consensus from evidence rather than authority. The formula is then constructed publicly on the board from the table columns, making the algebraic abstraction feel earned and logical rather than arbitrary. This embeds NGSS Science and Engineering Practice 8 (Obtaining, Evaluating, and Communicating Information) and Practice 5 (Using Mathematics and	OBSERVABLE INDICATORS: During Worked Example 1 (chapati, r=14, θ=90°), the correct answer is $A = (90/360) \times 3.14 \times 14^2 = 0.25 \times 3.14 \times 196 = 153.86$ cm². Teacher circulates and checks: (a) Did the learner square the radius BEFORE multiplying? (common error: multiplying r before squaring). (b) Did the learner correctly compute θ/360° as a fraction? (common error: computing 360/θ instead). Any learner with either error is redirected with: 'Show me just Column 4 from your table — what is πr^2? Now show me Column 5 — what fraction? Good. Now multiply

	annotate each symbol (θ = central angle in degrees, r = radius, $\pi \approx 3.14$ or 22/7), and work through two guided examples set in Kenyan contexts: (a) a round chapati of radius 14 cm cut at a central angle of 90°; (b) a shamba (farm plot) irrigated by a sprinkler that sweeps through an angle of 120° with a reach of 7 m.	What operation did we do next?' [Build the formula step-by-step from learner responses.] 4. Write on the board: $A = (\theta/360°) \times \pi r^2$. Say: 'This is not a formula I am giving you — this is a formula YOUR DATA BUILT. Read it back to me.' [Class reads aloud in unison.] 5. WORKED EXAMPLE 1: 'Mama Wanjiku bakes a large round chapati with radius 14 cm. She cuts one slice at a central angle of 90°. What is the area of that slice?' Think aloud each step. Then say: 'Now you do it in your book. Pause after each line and check with your neighbour.' [WAIT 4 minutes.] 6. WORKED EXAMPLE 2: 'A farmer in Nakuru uses a sprinkler that rotates through 120° and reaches 7 m. What area of the shamba is watered?' Cold-call one learner to work it on the board while others work in books. [WAIT 10 seconds before accepting a volunteer vs. cold-call.] 7. Explicitly connect back to phenomenon: 'Now go back to your Lesson 1 prediction. Was a slice with twice the angle exactly twice the area? How do you know with certainty now?'	Computational Thinking).	those two.' This reactivates the table scaffolding as a repair strategy.
Model Building Phase 📹 **VIDEO:** Proof: radius is perpendicular to a chord it bisects Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Learners individually write one new evidence card for the Driving Question Board: on one side they write 'WHAT WE OBSERVED' (one measurement finding from their data table) and on the other side 'WHAT IT MEANS' (the proportionality conclusion). Cards are posted on the DQB under the column 'Lesson 2 Evidence.' Learners then open their cumulative circle model (begun in Lesson 1 — a hand-drawn circle	1. Distribute evidence card slips (or direct learners to tear a page corner). Say: 'On one side write: WHAT WE OBSERVED — one specific piece of data from your table, with numbers. On the other: WHAT IT MEANS — one sentence about proportionality. You have 3 minutes.' [WAIT 3 minutes.] 2. Invite 3 learners to read their 'WHAT IT MEANS' sentence aloud before posting. Cold-call — do not accept volunteers only. [WAIT 10	CUMULATIVE MODEL REVISION ('Growing the Diagram'): Learners physically add to their ongoing circle diagram rather than starting a new page — making the accumulation of knowledge visible and tangible. The model functions as a personalised reference tool and a cognitive anchor. Each lesson's additions represent a measurable step forward in conceptual understanding. The DQB evidence card externalises	EXIT EVIDENCE: The one-sentence key inquiry answer functions as an exit ticket. Acceptable answer (target): 'We calculate the area of a sector by multiplying the fraction of the circle (θ/360°) by the full circle area (πr^2).' Partial understanding: 'We use the formula $A = (\theta/360°) \times \pi r^2$' (correct formula, no explanation of what it means). Insufficient: 'We measure the angle and radius' (describes the input, not the

Proportions with Angle Bisectors (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	diagram in their books) and add: (a) labelled parts — sector, segment, arc, central angle, chord; (b) the sector area formula A = (θ/360°) × πr² written inside the sector region; (c) an arrow labelled 'proportion of 360°' linking the central angle to the formula. Finally, learners write a one-sentence answer to the lesson's key inquiry question: 'How do we calculate the area of a part of a circle?'	seconds per learner.] 3. Direct learners to their cumulative model: 'Open to your circle model from Lesson 1. You are going to UPGRADE it with everything you learned today — labels, formula, and the proportion arrow. 4 minutes.' [WAIT 4 minutes. Circulate. Check that the formula is written inside the sector region.] 4. Close with the key inquiry question, called out to the whole class: 'Everyone, in one sentence — how do we calculate the area of a part of a circle? Write it. Now.' [WAIT 30 seconds.] Cold-call 2 learners to read their sentence. 5. Preview Lesson 3: 'In our next lesson, we will use this formula to solve more complex problems — including what happens when the angle is given in a different way. Keep your data tables safe.'	individual sensemaking into a communal knowledge artefact — learners can see how the class's collective understanding is building toward answering the driving question.	method). Teacher sorts sentences into three piles during break and uses them to plan targeted support at the start of Lesson 3.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. DATA QUALITY: Did learners measure the sector card radii and angles accurately enough (within ±0.5 cm and ±3°) to produce a clean proportional pattern in their tables? If not, what measurement technique needs more explicit modelling in future lessons — ruler placement? Protractor alignment?

2. PROPORTIONALITY UNDERSTANDING: When cold-called to explain whether a slice with twice the angle is twice the area, did learners use quantitative reasoning (citing specific numbers from their table) or qualitative reasoning ('it looks bigger')? The former is the target.

3. FORMULA OWNERSHIP: Did learners experience the formula $A = (\theta/360°) \times \pi r^2$ as something their data BUILT, or as something the teacher gave them? If the latter, the gallery walk and formula construction steps may need more time in a future iteration.

4. WORKED EXAMPLE ERRORS: What were the two most common errors in Worked Example 1 (chapati, $r=14$, $\theta=90°$)? Log these for targeted correction at the start of Lesson 3.

5. KENYA CONTEXT ENGAGEMENT: Did the Nakuru shamba sprinkler context generate genuine engagement and discussion, or did it feel forced? Could a more locally relevant context (e.g., a Maasai manyatta circular layout, or a round mutura cut at a market in Gikomba) produce deeper curiosity?

6. DQB QUALITY: Review the evidence cards posted on the Driving Question Board. Are learners connecting their measurements back to the phenomenon (the unequal chapati slices), or are they writing abstract mathematics? The DQB should feel like a story of growing understanding, not a vocabulary list.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students measured the radius and central angle of pre-cut sector card shapes (representing pizza/chapati slices of the same radius but different central angles: 45°, 90°, 135°, 180°). They recorded measurements in a structured data table and plotted a graph of central angle vs. sector area, observing a straight line through the origin.
What did I learn?	Students reviewed and labelled all parts of a circle (sector, segment, arc, chord, radius, diameter, central angle). They discovered through their own data that sector area is directly proportional to central angle — a slice with twice the angle has exactly twice the area — and connected this proportionality to the formula $A = (\theta/360°) \times \pi r^2$.
How does this explain the phenomenon?	Students explained that the area of a sector is calculated using $A = (\theta/360°) \times \pi r^2$, where θ is the central angle in degrees and r is the radius. They applied this formula to Kenyan contexts (a chapati slice of radius 14 cm at 90°, and a Nakuru shamba irrigated by a sprinkler sweeping 120° at 7 m reach), and updated their cumulative circle model with labelled parts and the formula.

LESSON 3 (40 minutes): EXPLAIN — Arc Length: Deriving the Formula Through the Fraction of a Circle	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students derive and apply the arc length formula by reasoning that a sector is a fraction of a full circle, then connect this to real Kenyan contexts including centre-pivot irrigation sweeps in the Rift Valley.
Knowledge	Students will know that arc length equals the fraction of the central angle out of 360 degrees multiplied by the full circumference, expressed as L = (theta/360) x 2pi r.
Skills	Students will be able to derive the arc length formula from first principles, substitute values of radius and central angle correctly, calculate arc lengths in real-world contexts, and express answers to appropriate degrees of accuracy.
Attitudes	Students will appreciate that mathematical formulas are not arbitrary rules but logical consequences of proportional reasoning, and will show curiosity about how geometry governs everyday Kenyan technology such as irrigation systems.
Key Inquiry Question	How do we calculate the arc length of part of a circle, and how does this connect to calculating the area of a sector?
Purpose in Storyline	Lesson 3 is the first EXPLAIN lesson. Students have already discovered in Lesson 2 that sector area is proportional to central angle. Now they ask: what about the curved edge of the slice — the arc? Deriving the arc length formula using the same proportionality logic builds the algebraic scaffolding they will need in Lesson 4 to derive the sector area formula. Arc length also appears naturally in the pizza phenomenon: the crust of each slice is an arc, and a bigger-angled slice has a longer crust — students can now calculate exactly how much longer.
Safety Notes	No significant safety concerns. If physical circular cut-outs or string are used for measurement, remind students to handle scissors or compasses carefully. Ensure adequate lighting for reading protractors.

B. LESSON OVERVIEW

Students begin by revisiting the pizza phenomenon: they notice the crust of each slice is a curved edge (an arc) and ask whether the same proportionality rule that governs area also governs arc length. Through guided inquiry students reason that if a full circle has circumference 2pi r, then a sector occupying theta degrees out of 360 degrees must have an arc that is (theta/360) of that circumference. They formalise this as L = (theta/360) x 2pi r, practise substituting values, and then apply the formula to a Kenyan centre-pivot irrigation context, calculating the arc swept by an irrigation boom across a circular shamba. The lesson ends with students updating the Driving Question Board with new evidence linking arc length and the upcoming sector area formula.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Circumference Principles - Basic] Source: CK-12	The teacher places the pizza/chapati cut-outs from Lesson 2 back on the desks. Students are asked to focus on the curved outer edge of each slice rather than its	Teacher holds up two cardboard pizza slices of visibly different angles and says: Yesterday we found that a slice with twice the angle has exactly twice the area.	Prediction-before-instruction activates prior knowledge of proportionality from Lesson 2 and creates cognitive investment. Comparing area proportionality to	Teacher listens for whether students spontaneously apply the proportionality reasoning from Lesson 2 to the new context of length. Students who are uncertain

🔍 Search ARES for similar videos 📖 **HTML:** Length Measurements to a Fraction of an Inch (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	flat interior. The teacher poses the question: If slice A has a central angle of 60 degrees and slice B has a central angle of 120 degrees, which slice has a longer crust — and by exactly how much? Students write individual predictions in their exercise books, expressing their thinking in their own words before any class discussion. They are then invited to share with a neighbour for 90 seconds.	Today I want us to think about the crust — the curved edge of the slice. Look at these two slices. Before we calculate anything, write in your book: which crust do you think is longer, and do you think the same doubling rule applies to length? [WAIT TIME: 60 seconds of silent individual writing.] Now turn to your partner and compare your predictions. [WAIT TIME: 90 seconds.] Teacher then cold-calls three pairs, probing with: What reasoning did you use to make that prediction? Are you assuming the same proportionality rule applies to length as it does to area? Why or why not?	length proportionality primes students to notice that the same fraction logic governs both.	whether proportionality applies to length reveal a conceptual gap that the Observe phase will address directly.
Observe Phase 📹 **VIDEO:** Circumference Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Length Measurements to a Fraction of an Inch (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students work in pairs with a piece of string, a ruler, and a cardboard circle of known radius (pre-cut in Lesson 2). They wrap the string around the full circumference of the circle, mark the length, then measure it with the ruler. They record the measured circumference and compare it to the calculated value 2pi r. Next, using the same string, they carefully trace only the arc of their assigned sector (central angle already recorded from Lesson 2) and measure that arc length. They record: central angle (theta), full circumference (2pi r), measured arc length, and the ratio (measured arc / full circumference). They complete a structured table in their books. A whole-class data collection is done on the board, gathering one row per pair.	Teacher circulates and asks: What fraction of the full circle does your sector represent? Can you write that fraction using your angle? [WAIT TIME: 30 seconds per pair before offering a hint.] When pairs have filled in the ratio column, teacher pauses the class and says: Look at the ratio column across all our rows. What do you notice? [WAIT TIME: 45 seconds silent observation.] Teacher then asks: How does that ratio relate to the central angle? Can you write it as a fraction with 360 in the denominator? Teacher writes the class data table on the board and invites a student to describe the pattern aloud, then asks the class: Does everyone agree? Can anyone offer a different interpretation?	Physical measurement with string grounds the abstract formula in tangible experience. The structured table makes the proportional pattern visible across multiple pairs, building collective evidence that arc length is always (theta/360) of the circumference — regardless of the specific angle.	Teacher checks the ratio column in students books as they work. Correct ratios should equal theta/360. Students who record ratios that do not simplify to theta/360 are guided to re-examine their string measurement technique or their angle reading from the protractor.
Explain Phase 📹 **VIDEO:** Circumference Principles - Basic	Building on the observed pattern, students are guided to derive the formula step by step. The teacher writes the proportionality statement	Teacher writes on the board: We have seen from our data that arc/circumference = theta/360. Now let us write this as an	Deriving the formula from the proportion statement (rather than simply receiving it) ensures students understand its logical	Teacher collects answers to all three examples. Correct answers: Example 1 — approximately 17.6 cm; Example 2 — approximately

Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Length Measurements to a Fraction of an Inch (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	on the board: arc length is to full circumference as theta is to 360 degrees. Students copy and then individually complete the algebraic step to isolate arc length: L/2pi r = theta/360, therefore L = (theta/360) x 2pi r. Students then work through three graded examples in their books. Example 1: A pizza of radius 14 cm is cut at 72 degrees — find the arc length of that slice (the crust). Example 2: A fan-shaped piece of land has a radius of 50 m and a central angle of 150 degrees — find the arc length. Example 3 (Kenyan context — full narrative): A centre-pivot irrigation system at a wheat farm in Nakuru County has a boom of length 120 m. It sweeps through an angle of 210 degrees overnight. Calculate the arc length traced by the tip of the boom, giving the answer in metres correct to one decimal place. Students solve independently, then compare answers with a partner before a class debrief.	equation and solve for arc length. [Teacher writes the proportion and pauses.] Before I simplify, can someone tell me what we must do to both sides to isolate L? [WAIT TIME: 45 seconds.] Teacher cold-calls a student and affirms or corrects the algebraic step. Teacher then says: Write the final formula in a box in your book. This is a formula you have derived yourself — it is not magic, it is proportionality. For Example 3, teacher sets the scene: Imagine you are an engineer at a large wheat farm near Lake Nakuru. Your irrigation boom is 120 metres long. It sweeps 210 degrees overnight. The farmer needs to know how much fence to build along the arc to protect the watered crop from wildlife — that is the arc length you are calculating. [WAIT TIME: 3 minutes for independent work on Example 3.] Teacher circulates, looking for errors in unit conversion or misuse of diameter instead of radius. After the debrief, teacher asks: How does this arc length formula connect to what we found about sector area in Lesson 2? What do you predict the sector area formula will look like?	basis. Contextualising Example 3 in a named Kenyan county (Nakuru) and a real agricultural technology (centre-pivot irrigation) makes the mathematics purposeful. The final predictive question bridges to Lesson 4, advancing the driving question storyline.	130.9 m; Example 3 — approximately 439.8 m. Common errors to watch for: using diameter instead of radius; writing the formula as (theta/180) x pi r (radian confusion); forgetting to multiply by 2 in 2pi r. Teacher addresses these errors publicly during the debrief using a think-aloud correction model.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Circumference Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Length	Students return to the class Driving Question Board. They write new sticky notes responding to two prompts: (1) New evidence we now have — what does arc length tell us about a pizza slice that we did not know before? (2) New questions this raises — if arc length uses the formula (theta/360) x 2pi r, what do you predict the formula for the area of a sector will look like, and why? Students place their notes on the DQB under the	Teacher says: In Lesson 1 you predicted how area relates to angle. In Lesson 2 you confirmed that prediction with data. Today you have derived a formula for arc length. How does having the arc length formula change or deepen your understanding of the pizza slicing phenomenon? [WAIT TIME: 60 seconds for students to think and write their sticky notes.] Teacher reads two or three notes aloud and probes: This student	The DQB update makes student thinking visible and cumulative. By explicitly comparing the arc length formula structure to the anticipated sector area formula, the teacher activates forward-looking reasoning and creates a productive intellectual gap that motivates Lesson 4.	Teacher reads the NEW EVIDENCE sticky notes to check whether students have correctly identified that arc length is proportional to angle and to radius. Notes that misstate the relationship (for example claiming arc length depends only on angle, not radius) are addressed gently in whole-class discussion.

Measurements to a Fraction of an Inch (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	headings NEW EVIDENCE and NEW QUESTIONS. The teacher reads several notes aloud and the class discusses whether the new evidence supports or complicates any earlier predictions posted in Lesson 1.	says the arc formula looks almost like the area formula. What is the same? What is different? What does that similarity suggest to you about how area of a sector might be calculated? Teacher leaves the NEW QUESTIONS column visibly incomplete, signalling that Lesson 4 will supply the answer.		
Model Building Phase 📹 **VIDEO:** Circumference Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Length Measurements to a Fraction of an Inch (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students update their personal mathematical model in their exercise books. They draw a clearly labelled diagram of a sector showing radius r, central angle theta, and arc length L. They write the arc length formula beneath the diagram, annotating each term (theta divided by 360 gives the fraction of the circle; 2pi r is the full circumference; multiplying gives the arc length). Students then write a one-sentence connection statement: The arc length formula and the sector area formula both start with theta/360 because both express a fraction of a full circle — one for length, one for area. Finally, students solve one exit-ticket problem independently: A circular shamba in Kisumu County has a radius of 35 m. A footpath follows the arc of a minor sector with a central angle of 80 degrees. How long is the footpath? Students submit this on a half-sheet of paper as they leave.	Teacher says: Before we close, I want you to build a model — a diagram plus a formula plus a sentence — that captures everything you have understood today. This model should be clear enough that you could use it to teach a classmate who was absent. [WAIT TIME: 4 minutes for model drawing and writing.] Teacher circulates and looks for correct labelling and the full formula. Teacher then says: For the exit ticket, work independently and silently. Show every step. We will use these to start Lesson 4. [WAIT TIME: 3 minutes.] Teacher collects exit tickets. Before students leave, teacher says: Tomorrow we will use the exact same fraction reasoning to derive the area of a sector. Keep that question in your mind overnight: if arc length is (theta/360) x 2pi r, what do you expect sector area to be?	The annotated diagram forces students to connect symbolic notation to geometric meaning. The connection sentence requires integration of todays learning with Lesson 2. The exit ticket provides immediate formative data on formula application accuracy before the next lesson.	Exit ticket correct answer: L = (80/360) x 2 x pi x 35 = approximately 48.9 m. Teacher sorts exit tickets into three piles: correct with full working, correct answer with incomplete working, and incorrect. The incorrect pile informs the opening review of Lesson 4. Students who used diameter (70 m) instead of radius are the most common error group and will receive targeted correction at the start of the next lesson.

D. TEACHER REFLECTION

Did students derive the formula independently or did they require heavy scaffolding? Were students able to articulate the fraction reasoning in their own words, or did they treat the formula as a rule to memorise? Did the Nakuru irrigation context feel authentic and engaging, or did it seem abstract? Review exit ticket piles: if more than one-third of students made errors, begin Lesson 4 with a five-minute re-teaching of the proportion derivation before introducing sector area. Consider whether any student showed the insight that arc length and sector area share the same fractional structure — celebrate that insight publicly at the start of Lesson 4 to reward and model mathematical reasoning.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students measured arc lengths of sectors using string and rulers, then collected class data showing that the ratio of arc length to full circumference always equals theta divided by 360, confirming that arc length is directly proportional to central angle.
What did I learn?	Students derived the arc length formula L = (theta/360) x 2pi r from a proportionality statement, applied it to three graded examples including a centre-pivot irrigation sweep in Nakuru County, and expressed answers to appropriate accuracy.
How does this explain the phenomenon?	The arc of a sector is a fraction of the full circumference, and that fraction equals the central angle divided by 360 degrees. This same fractional reasoning will govern the sector area formula in Lesson 4, because both arc length and sector area represent proportional parts of their respective whole-circle quantities.

LESSON 4 (80 minutes): EXPLAIN — Deriving and Applying the Area of a Sector (Minor Sector)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To derive the formula for the area of a sector and apply it to solve real-life problems in Kenyan contexts including food, farming, and design.
Knowledge	By the end of the lesson, learners should be able to: (a) state the formula for the area of a sector as $A = (\theta/360°) \times \pi r^2$; (b) explain why the sector area formula is analogous to the arc length formula; (c) calculate the area of a minor sector given the radius and central angle; (d) distinguish between a sector and a segment and use both areas to solve problems.
Skills	By the end of the lesson, learners should be able to: (a) derive the sector area formula by reasoning from proportionality; (b) substitute values of θ and r correctly into $A = (\theta/360°) \times \pi r^2$; (c) apply the formula to real-world problems involving a pizza slice, a maize shamba, and a bead panel design; (d) verify computed answers using a scientific calculator.
Attitudes	Learners appreciate how a single mathematical formula connects measurement, proportional reasoning, and real-life problem solving in Kenyan agriculture, food, and design contexts.
Key Inquiry Question	1. How do we calculate the area of a part of a circle? 2. How do areas of sectors and segments of a circle apply in day-to-day activities?
Purpose in Storyline	Lessons 1–3 established the phenomenon (unequal pizza/chapati slices), introduced arc length, and generated the Driving Question Board. Lesson 4 is the EXPLAIN phase: learners now derive the sector area formula by analogy with arc length, apply it to the original pizza phenomenon to confirm or refute their earlier predictions, and extend it to a maize shamba and a Maasai bead panel — advancing the driving question toward a complete, generalisable answer.
Safety Notes	No significant hazards. If a real pizza or chapati is brought in, ensure food hygiene protocols are observed. Remind learners not to share food if any allergies exist. Handle protractors and compasses with care to avoid injury.

B. LESSON OVERVIEW
Lesson 4 is the formal EXPLAIN lesson in the evidence-gathering sequence. Building on the arc length derivation from Lesson 3, learners use proportional reasoning to derive the sector area formula $A = (\theta/360°) \times \pi r^2$. The anchoring phenomenon — unequal slices of a circular pizza/chapati — is revisited: learners now have the tool to confirm whether a slice with twice the central angle is exactly twice the area. Three graded application problems (pizza slice, maize shamba sector, Maasai bead panel) move from the familiar food context to farming and design. Learners work in pairs, verify with calculators, and update the class Driving Question Board. The lesson closes with a model-building revision that layers the sector area formula onto the circle diagram started in Lesson 1.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Video: Circuit Construction Kit:]	Learners are shown the same circular pizza/chapati diagram from Lesson 1, now annotated with two slices: Slice A (central angle 60°,	Display the annotated pizza diagram on the board (drawn or projected). Say: 'Look carefully at these two slices from our pizza.	Strategy: Prediction with Justification. Learners activate prior knowledge of proportionality and circle area (πr^2) before formal	Observable indicator: Each learner has a written prediction AND a written reason in their exercise book before partner talk begins.

DC Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Quadratic Formula (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	radius 14 cm) and Slice B (central angle 120°, radius 14 cm). Working individually in their exercise books for 3 minutes, they write a prediction in response to the prompt: 'Slice B has exactly twice the central angle of Slice A. Predict whether Slice B has exactly twice the area. Give a reason for your prediction using what you already know about circles.' Learners then share predictions with their shoulder partner for 2 minutes before the teacher cold-calls 4 learners to share with the class. Predictions are recorded on a sticky-note strip on the Driving Question Board under the column 'Our Predictions — Lesson 4'.	Both come from the same pizza — same radius, 14 centimetres. But Slice B has an angle of 120 degrees and Slice A has an angle of 60 degrees. In your book, predict: is the area of Slice B exactly twice the area of Slice A? Write your reason.' Allow 3 minutes of silent individual thinking — do NOT answer clarifying questions yet. After 3 minutes say: 'Turn to your shoulder partner. You have 2 minutes to compare your predictions.' After partner talk, cold-call exactly 4 learners (choose 1 high-confidence, 2 mid-confidence, 1 who looks uncertain). For each response ask: 'What in your previous learning makes you say that?' Record each prediction on the DQB strip verbatim. Do not confirm or deny any prediction. Close with: 'By the end of today's lesson, you will be able to calculate both areas exactly and settle this question once and for all.'	instruction. Requiring a written reason forces retrieval of relevant prior knowledge (arc length proportionality from Lesson 3) and creates cognitive dissonance that motivates the derivation. Recording predictions publicly on the DQB creates accountability and makes the lesson's resolution feel earned.	Teacher circulates during the 3-minute silent period and scans for: (a) learners who write 'yes, twice' with a proportionality reason — these learners are on track; (b) learners who write 'no' without a reason — flag for targeted questioning during Phase 2; (c) learners who leave the reason blank — provide the prompt 'Think about how arc length was related to the full circumference in Lesson 3.'
Observe Phase 📹 **VIDEO:** Video: Circuit Construction Kit: DC Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Quadratic Formula (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their notebooks to the arc length derivation notes from Lesson 3. The teacher guides a whole-class structured derivation using the 'Analogy Bridge' method — a two-column table drawn on the board headed 'Arc Length (what we already know)' and 'Sector Area (what we are discovering)'. Row by row, learners fill in the table in their own books alongside the teacher. The table has five rows: (1) Full circle measurement [Circumference = $2\pi r$ | Area = πr^2]; (2) Fraction of the circle [θ/360° | θ/360°]; (3) Formula [Arc length = (θ/360°) × $2\pi r$ | Sector area = (θ/360°) × πr^2]; (4) What it means in words [The arc is that fraction of the full	Draw the two-column Analogy Bridge table on the board. Say: 'In Lesson 3 we derived the arc length formula by asking: what fraction of the full circle is our sector? Today we use exactly the same logic — but for AREA instead of length.' Complete Row 1 together: ask 'What is the area of the FULL circle?' — wait 10 seconds — cold-call 1 learner. Write πr^2 in the right column. For Row 2 say: 'What fraction of the full 360 degrees does our sector take up?' — wait 10 seconds — cold-call 1 learner. Write θ/360° in both columns. For Row 3 say: 'So if the full area is πr^2 and our sector is θ/360° of the full circle, write the formula for sector area in your book. Do it now — do	Strategy: Analogy Bridge (Structured Comparison Table). By placing arc length and sector area side by side in matching rows, learners see that the same proportional logic governs both formulas. This reduces cognitive load (the derivation is not new reasoning — it is the same reasoning applied to area instead of length) and deepens understanding of why the formula works rather than just what it is. Completing the table in their own books ensures active encoding rather than passive copying.	Observable indicator: After the Pizza Check, every learner should have A = 205.2 cm² for Slice B written with at least three visible steps: (1) substitution, (2) fraction simplification, (3) multiplication by π×196. Teacher circulates during the 3-minute independent calculation and looks for: (a) learners who get 205.25 cm² — correct, praise precision; (b) learners who calculate πr^2 first then multiply by the fraction — equally valid, affirm; (c) learners who forget to square r (writing π×14 instead of π×14²) — this is the most common error; address immediately by asking 'What is the area of the FULL pizza? Can it be smaller than our slice?'

	circumference | The sector is that fraction of the full area]; (5) Pizza check — Slice A, $\theta=60°$, $r=14$ cm [Arc = $(60/360)\times 2\pi\times 14 = 14.67$ cm | Area = $(60/360)\times\pi\times 14^2 = 102.6$ cm^2]. Learners then calculate Slice B independently and compare with their Lesson-4 prediction on the DQB.	not look at your partner's book.' Wait 15 seconds. Cold-call 2 learners and write $A = (\theta/360°) \times \pi r^2$ on the board. Underline it. Say: 'This is the formula. Write it in a box in your book.' For the Pizza Check (Row 5), say: 'Use your calculator. Find the area of Slice A: $\theta = 60°$, $r = 14$ cm. Show every step.' Wait 2 minutes. Cold-call 1 learner to read out their steps. Correct any errors on the board. Then say: 'Now calculate Slice B on your own — $\theta = 120°$, $r = 14$ cm. You have 3 minutes.' After 3 minutes cold-call 1 learner. Write the answer (205.2 cm^2) on the board. Say: 'Look at your DQB prediction. Were you right? Is 205.2 exactly twice 102.6?' — pause 5 seconds for mental arithmetic — 'Yes! Exactly twice. The relationship IS perfectly proportional. Why? Because the only thing that changed was θ, and θ appears as a simple multiplier.'		
Explain Phase 📹 **VIDEO:** Video: Circuit Construction Kit: DC Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Quadratic Formula (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in pairs on three graded application problems, each set in a different Kenyan context. The problems are written on the board or on a printed/projected worksheet. Problem 1 (Pizza Phenomenon — direct connection): A round chapati at a Nairobi café has radius 15 cm. It is cut into a slice with a central angle of 80°. Calculate the area of the slice. Problem 2 (Maize Shamba Sector — farming context): A farmer in the Rift Valley irrigates a circular plot of land using a pivot irrigation system. The pivot arm is 21 metres long (radius = 21 m). One morning, the pivot arm sweeps through an angle of 135°. What is the area of the maize shamba watered that morning?	Say: 'You now have the formula. Let's use it. I have three problems — each one connects the formula to a real situation in Kenya. Work in pairs. Show all working. You have 15 minutes.' Write the three problems on the board clearly, labelling θ and r for Problem 1 as a scaffold ($\theta = 80°$, $r = 15$ cm). Do NOT scaffold θ and r for Problems 2 and 3 — learners must identify these from the problem text. Circulate actively. At the 5-minute mark, check that all pairs have completed Problem 1 correctly ($A \approx 157.1$ cm^2). For pairs still on Problem 1, ask: 'What is θ? What is r? Substitute.' For Problem 2, listen for confusion about 'pivot arm = radius'. If heard, ask the whole class: 'In the irrigation	Strategy: Graduated Contextual Application (Near–Far Transfer). Problem 1 is a near transfer (same food context as the phenomenon). Problem 2 is a mid-range transfer (agriculture — different domain but same geometry). Problem 3 is a far transfer (design and exact values — requires algebraic rather than decimal output). This gradation builds fluency and generalisation systematically. Pair work followed by group reconciliation exposes misconceptions through peer dialogue before teacher correction, which research shows produces deeper retention than teacher-led correction alone.	Observable indicators for each problem: P1 — learner writes $A = (80/360) \times \pi \times 15^2$, simplifies to $(2/9) \times 225\pi$, arrives at 157.1 cm^2 (3 s.f.); P2 — learner correctly identifies $r = 21$ m and $\theta = 135°$, writes $A = (135/360) \times \pi \times 441$, arrives at 1,039 m^2; P3 — learner writes $A = (72/360) \times \pi \times 64 = (1/5) \times 64\pi = 64\pi/5$ cm^2 and does NOT convert to a decimal. Common errors to watch for: (a) using diameter instead of radius ($r = 30$ instead of 15 in P1); (b) not squaring r; (c) failing to simplify the fraction $\theta/360°$ before multiplying. Any learner making error (a) or (b) must redo the calculation with teacher guidance before moving to Phase 4.

	Give your answer to the nearest square metre. Problem 3 (Maasai Bead Panel — design context): A Maasai beadwork panel is designed around a circle of radius 8 cm. A minor sector with central angle 72° is filled with red beads. A craftsperson wants to know the exact area to be covered in red. Calculate the exact area, leaving your answer in terms of π. After completing all three, pairs compare answers with another pair (group of 4) and reconcile any differences before the teacher reveals solutions.	problem, what does the pivot arm represent geometrically?' — wait 10 seconds — cold-call 1 learner. At 15 minutes, say: 'Stop. Compare your answers with the pair next to you. You have 3 minutes to agree on a single set of answers.' After 3 minutes, cold-call one group of 4 to give each answer. Write the correct solutions on the board: P1 = 157.1 cm², P2 = 1,039 m² (nearest m²), P3 = 16π/5 cm² (exact). For Problem 3, say: 'Why did I ask for the answer in terms of π? Because it is EXACT. 16π/5 is more precise than any decimal. A craftsperson cutting beads to the millimetre would prefer this.' Connect back to the phenomenon: 'Every one of these problems is the same as our pizza slice — different radius, different angle, same formula.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Circuit Construction Kit: DC](#) Source: PhET Interactive Simulations (English) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Quadratic Formula (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	The class pauses for a 10-minute structured DQB update. Learners return to the class DQB (a large wall display or whiteboard section maintained since Lesson 1). Working individually for 3 minutes, each learner writes two things on separate sticky notes: (1) a GREEN note — one thing today's lesson has now answered about the driving question ('How can we use the radius and the central angle to calculate the exact area of a pizza slice?'); (2) an ORANGE note — one new question or confusion that today's lesson has raised (e.g., 'What if the slice is less than a triangle? How do we find the area of the curved segment between the chord and the arc?'). Learners post their notes on the DQB. The teacher reads selected orange notes aloud and previews Lesson 5: 'Your	Say: 'Before we build our model, we update our Driving Question Board — the record of what we have figured out together.' Distribute two sticky notes per learner (green and orange, or label plain notes G and O). Say: 'Green note: write one specific thing you can now DO that answers part of our driving question. Orange note: write one new question or one thing that still puzzles you about circular areas.' Allow 3 minutes of silent individual writing. Circulate and read over shoulders — prompt any learner with a blank orange note: 'Think about the sliver between the chord and the arc of our pizza slice — is that a sector? What would you call it?' After posting, read 3 green notes and 3 orange notes aloud. For each green note say: 'Good — you can now [restate the learner's point].	Strategy: Driving Question Board — Evidence Accumulation Tracking. The DQB is a metacognitive tool that makes the growth of understanding visible over the 10-lesson sequence. Green notes consolidate and celebrate genuine progress; orange notes preserve productive curiosity and drive the next lesson's inquiry. Publicly posting contributions holds the class community accountable to the shared inquiry and signals that all questions are valued evidence of thinking.	Observable indicator: Every learner posts at least one green note that includes the formula A = (θ/360°) × πr² or a correct worked example, AND one orange note that identifies a genuine remaining question (not a repetition of something already answered). Teacher scans all notes after class. Any learner whose green note is incorrect (e.g., states the wrong formula) is flagged for a brief one-on-one check at the start of Lesson 5.

	orange notes tell me exactly what we explore next — the area of a segment. You are building the map of this unit yourselves.'	That IS part of our answer.' For each orange note say: 'Excellent question. Hold it — it becomes the focus of Lesson 5.' Do NOT answer the orange-note questions now. Close this phase by saying: 'Our driving question asked how we use radius and central angle to get exact area. You now have the formula. You have used it in three real Kenyan problems. The driving question is almost answered — almost.'		
Model Building Phase 📹 **VIDEO:** [Video: Circuit Construction Kit: DC] Source: PhET Interactive Simulations (English) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Quadratic Formula (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Learners retrieve the cumulative annotated circle diagram they have been building since Lesson 1 (held in a dedicated page of their exercise book or on a printed template). Today they add two new layers to their model: (1) They label a sector with angle θ and radius r, and write the formula $A = (\theta/360°) \times \pi r^2$ inside the sector with an arrow pointing to the formula. (2) They draw a chord across the sector to create a segment, label it, and write 'Area of segment = Area of sector − Area of triangle' as a 'coming next' annotation. Learners also add a real-world icon next to the sector label: a small sketch of a pizza slice (phenomenon), a pivot irrigation arc (farming), or a bead panel wedge (design) — their choice. The model is shared briefly with a partner who checks that: (a) the formula is correctly written; (b) the radius and angle are labelled; (c) the segment annotation is present. Partners sign each other's model diagram as a peer-review stamp.	Say: 'Take out your cumulative circle model — the diagram you have been adding to since Lesson 1. Today we add the sector area layer.' Project or draw a reference version of the model on the board showing today's additions. Say: 'Step 1: Inside your sector, write the formula and draw an arrow from the sector to the formula. Step 2: Draw a chord — a straight line — connecting the two radius endpoints of your sector. Label the region between the chord and the arc as a segment. Write: Area of segment = Area of sector − Area of triangle. This is a PREVIEW — we derive it in Lesson 5.' Allow 7 minutes for independent model updating. Then say: 'Swap your book with your partner. Check: Is the formula written correctly? Are r and θ labelled? Is the segment annotation there? Sign your partner's diagram.' Allow 3 minutes for peer review. Collect 5 books at random to review tonight as a teacher formative check. Close the lesson by returning to the phenomenon: 'We started this unit with a pizza cut into unequal slices and a question: is a slice with twice the angle always exactly twice the area? Today you proved:	Strategy: Cumulative Model Building with Peer Review. The annotated diagram is a running visual knowledge artefact that grows across all 10 lessons. Each addition forces learners to integrate new knowledge with prior layers (arc length, circumference, full circle area). Adding the 'coming next — segment' annotation bridges to Lesson 5 and preserves the spirit of inquiry. Peer review with a signature makes the check authentic and social — learners are more careful when a peer will inspect their work.	Observable indicator: Teacher's random collection of 5 books after the lesson. Success criteria for the model: (1) $A = (\theta/360°) \times \pi r^2$ is written correctly (not $A = \theta/360 \times \pi r$ or similar errors); (2) θ and r are both labelled on the diagram; (3) a chord is drawn and the segment is labelled; (4) the segment area annotation reads 'Area of sector − Area of triangle' (not just 'segment area'). Any book failing criteria (1) or (2) triggers a targeted 5-minute reteach at the opening of Lesson 5.

		YES — and you have the formula to calculate the exact area of any slice, any shamba, any bead panel. That is real mathematical power.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. FORMULA DERIVATION — Did the Analogy Bridge table help learners see WHY the formula works, or did most learners treat it as a rule to memorise? Evidence: listen to the language learners use during Phase 3 problem solving — do they say 'I'm finding what fraction of the full area this sector is' (conceptual) or 'I'm just using the formula' (procedural)? If mostly procedural, plan a brief re-derivation at the start of Lesson 5 using a different analogy (e.g., fraction of a githeri serving).

2. COMMON ERRORS — Which of the three predicted errors (using diameter instead of radius, not squaring r, failing to simplify θ/360°) appeared most frequently? Tally from your Phase 3 circulation notes and adjust the opening of Lesson 5 accordingly.

3. DQB ORANGE NOTES — Do learners' orange notes show that the majority are asking about segments (the intended bridge to Lesson 5), or are some still confused about the basic sector formula? If more than 25% of orange notes reveal sector confusion, insert a 10-minute consolidation activity at the start of Lesson 5 before introducing segments.

4. KENYA CONTEXTS — Did learners engage more strongly with the pizza/chapati problem, the irrigation shamba problem, or the bead panel problem? Note which context generated the most spontaneous discussion — use that context as the anchor for Lesson 5's segment derivation.

5. PACE — Was 15 minutes sufficient for the three application problems in Phase 3? If most pairs did not reach Problem 3, consider assigning it as a homework extension and opening Lesson 5 with a brief whole-class solution of Problem 3 before the segment derivation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that two pizza/chapati slices from the same circular food (radius 14 cm) with central angles 60° and 120° have areas of 102.6 cm^2 and 205.2 cm^2 respectively — confirming that doubling the central angle exactly doubles the sector area, making the relationship perfectly proportional.
What did I learn?	Learners derived and applied the sector area formula $A = (\theta/360°) \times \pi r^2$ by analogy with the arc length formula, understanding that a sector is simply a proportional fraction (θ/360°) of the full circle's area (πr^2).
How does this explain the phenomenon?	Learners explained that the formula works because any sector is a fraction of the full circle determined solely by its central angle, and that the same proportional logic that gives arc length also gives sector area — enabling exact calculation of the area of any pizza slice, maize shamba sector, or bead panel wedge given only the radius and central angle.

LESSON 5 (40 minutes): EXPLAIN — Area of a Major Sector	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To extend the sector area formula to major sectors (central angle greater than 180 degrees) and apply it to real Kenyan contexts including irrigation fields and matatu windscreen wipers.
Knowledge	Learners will know that a major sector has a central angle greater than 180 degrees and that the same formula A = (theta divided by 360) multiplied by pi r squared applies to both minor and major sectors; they will also know that the areas of the minor and major sector of the same circle always sum to the full circle area pi r squared.
Skills	Learners will be able to identify and distinguish between minor and major sectors from a diagram or a described situation, calculate the area of a major sector using the formula, and verify their answer by checking that minor sector area plus major sector area equals pi r squared.
Attitudes	Learners will appreciate that a single unified formula can describe both small and large portions of a circle, building confidence in the power of proportional reasoning and mathematical generalisation.
Key Inquiry Question	How do we calculate the area of a part of a circle — specifically a part that is larger than a half-circle?
Purpose in Storyline	In Lesson 4 learners could calculate the area of any pizza slice narrower than a half-circle. Now they confront the complementary question: what is the area of the REMAINING pizza after a minor slice is removed? This is the major sector. Resolving this extends their formula toolkit and advances the Driving Question Board by showing that one formula handles all sector sizes.
Safety Notes	No physical cutting tools are used in this lesson. Compasses used for diagrams should be handled with care. Ensure learners cap compasses when not in use.

B. LESSON OVERVIEW

Learners begin by revisiting the pizza phenomenon: after a narrow slice (minor sector) is removed, they predict the area of the large remaining piece. They observe that the remaining angle is greater than 180 degrees and must decide whether their formula still works. Through guided inquiry they confirm that A = (theta divided by 360) multiplied by pi r squared is universal, apply it to a major sector of a centre-pivot irrigation field (the unwatered portion) and to the sweep of a matatu windscreen wiper, and verify that minor plus major sector areas equal the full circle. The lesson closes with an updated Driving Question Board entry and a revised sector model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [More on Normal force (shoe on floor)] Source: Khan	The teacher displays a large paper circle (representing the pizza from Lesson 1) from which a 60-degree minor sector has already been cut out and set aside. Learners look at the large remaining piece. In their	Hold up the large remaining piece and ask: 'We calculated the small slice in our last lesson. Now look at this big leftover piece — what do we call it, and how would we find its area?' WAIT 30 SECONDS	Eliciting prior knowledge — surfacing intuitive ideas about complementary areas and fractions of a circle before formal vocabulary is introduced.	Listen for whether learners naturally subtract the minor sector area from the full circle area (informal complement reasoning) versus those who are unsure how to begin. This reveals readiness

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Proportions with Angle Bisectors (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	exercise books they write a prediction: (a) Estimate — what fraction of the whole pizza is the remaining piece? (b) Predict — if the original pizza had radius 15 cm, what is the approximate area of the remaining piece? Learners share predictions with a shoulder partner, then three pairs share with the class. Predictions are recorded on the board without correction.	for individual thinking before allowing partner talk. Probe with: 'Is this piece more or less than half the circle? How do you know?' Record all numerical estimates on the board. Resist confirming or rejecting any prediction at this stage.		for the formal derivation.
Observe Phase 📹 **VIDEO:** More on Normal force (shoe on floor) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Proportions with Angle Bisectors (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a printed worksheet showing three circles, each with a shaded sector: Circle 1 has a 60-degree shaded sector (minor), Circle 2 has a 200-degree shaded sector (major), Circle 3 has a 270-degree shaded sector (major). For each circle, learners (i) label the shaded sector as minor or major, (ii) write the angle of the UNSHADED sector, (iii) confirm that shaded angle plus unshaded angle equals 360 degrees. They also measure the angles of the major sectors in Circles 2 and 3 with a protractor to confirm the printed values. Groups then discuss: 'What is the same about all three sector area calculations? What is different?'	Circulate and ask targeted questions: 'How did you decide which sector is major?' and 'If I give you only the minor angle, how can you find the major angle?' WAIT 20 SECONDS after each question. After groups settle, ask one group to explain how they found the unshaded angles. Introduce the vocabulary REFLEX ANGLE (greater than 180 degrees) and MAJOR SECTOR formally at this point, writing definitions on the board.	Structured observation with a comparison table — learners move from concrete visual discrimination (shaded vs unshaded) to noticing that 360-degree symmetry governs both sectors, setting up the algebraic generalisation.	Check completed worksheets for correct labelling and complementary angle pairs. Misconception to watch for: learners who think the formula changes for angles above 180 degrees.
Explain Phase 📹 **VIDEO:** More on Normal force (shoe on floor) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Proportions with Angle Bisectors (Flexbook) Source: CK-12	Step 1 — Derive: The teacher guides learners to substitute theta = 200 degrees and r = 15 cm into A = (theta divided by 360) multiplied by pi r squared. Learners calculate on paper, then verify by also computing the minor sector (theta = 160 degrees) and checking the two areas sum to pi times 15 squared. Step 2 — Kenyan Context A (Irrigation Field): A centre-pivot irrigation system in Laikipia has radius 80 m. It waters a minor sector of 110 degrees. Learners calculate (i) the	Step 1: Write the formula on the board and say: 'Before we calculate anything new, let us test whether our formula dares to accept an angle bigger than 180. Try theta = 200 degrees.' WAIT 2 MINUTES for individual calculation. Step 2: After checking answers, say: 'Now let us visit a real shamba in Laikipia. A farmer tells us the pivot waters 110 degrees — but the county agricultural officer wants to know how much land is NOT being watered. What do we do first?'	Worked example progression — learners move from a numerical test case (theta = 200) to two authentic Kenyan problems, then to a limiting-case verification (theta = 360), building confidence that the formula is universal. Peer checking of the complement verification (minor plus major = full circle) acts as built-in error detection.	Circulate during Step 2 and check that learners correctly identify the major sector angle as 360 minus 110 = 250 degrees before substituting. Common error: using 110 degrees instead of 250 degrees for the unwatered area. Ask learners who finish early to design their own major-sector problem set in a Kenyan context of their choice.

🔍 Search ARES for similar readings	watered area and (ii) the UNwatered major sector area (250 degrees). Step 3 — Kenyan Context B (Matatu Wiper): A matatu windscreen wiper sweeps a major arc. The wiper arm is 28 cm long, and the wiper blade sweeps through a reflex angle of 210 degrees on its return stroke. Learners calculate the area cleaned on the return stroke. Step 4 — Generalisation: Learners write in their own words: 'The formula A = (theta divided by 360) multiplied by pi r squared works for ANY sector because theta divided by 360 simply gives the fraction of the full circle, whether that fraction is small or large.'	WAIT 30 SECONDS. Step 3: Introduce the matatu context with a quick sketch on the board of a wiper arc. Ask: 'Why might an engineer designing windscreen wipers need to calculate this area?' WAIT 20 SECONDS. Step 4: Prompt the generalisation by asking: 'What would happen to our formula if theta were exactly 360 degrees? What answer would we get, and does that make sense?' WAIT 30 SECONDS — expected answer: the full circle area pi r squared.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** More on Normal force (shoe on floor) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Proportions with Angle Bisectors (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group reviews the Driving Question Board from previous lessons. They add one NEW sticky note in the EVIDENCE column answering: 'We now know that the formula A = (theta divided by 360) multiplied by pi r squared calculates the area of BOTH minor and major pizza slices — and that the two slices always add up to the full pizza area.' They also move any earlier question sticky notes that this lesson has answered into the ANSWERED section, and post any new questions that arose (for example: 'What if the sector boundary is a chord instead of two radii — is that still a sector?').	Say: 'Go to the DQB. Find a question from an earlier lesson that today has helped answer. Move it to the ANSWERED column and write a one-sentence answer beside it.' WAIT 3 MINUTES for group work at the board. Then invite one group to read their new evidence note aloud. Ask the class: 'Does this evidence change or confirm our earlier predictions about the pizza?' WAIT 20 SECONDS.	DQB as a living knowledge-tracking tool — physically moving sticky notes from QUESTIONS to ANSWERED makes the progression of inquiry visible and reinforces that evidence accumulates across lessons.	Observe which earlier questions groups choose to answer — this reveals whether they are connecting today's learning to the broader inquiry. Listen for the distinction between sector and segment in any new questions posted, which will preview Lesson 6.
Model Building Phase 📹 **VIDEO:** More on Normal force (shoe on floor)	Learners update their circle-parts model (begun in Lesson 2 and extended each lesson). They add a new diagram showing a circle divided into a minor sector (theta degrees) and a major sector (360 minus theta degrees), label both	Say: 'Add today's discovery to your model. Draw both sectors clearly and label both angles. Make sure your model shows the relationship between the two sector areas.' Circulate and prompt with: 'If I cover up the formula and show	Cumulative model building — the running diagram and formula summary table serve as an external memory aid that makes all prior learning visible in one place, supporting transfer and revision in Lessons 8 and 9.	Exit prompt responses reveal two things: (i) whether learners can correctly compute the Laikipia area (expected: approximately 13 962 square metres for 250 degrees and r = 80 m), and (ii) whether their justification references the

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Proportions with Angle Bisectors (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	with their area formulas, and draw a double-headed arrow labelled 'together they equal pi r squared.' They then complete a three-row summary table in their books: Row 1 — Arc length formula; Row 2 — Minor sector area; Row 3 — Major sector area. Finally, learners answer a two-sentence exit prompt: 'The area of the unwatered portion of the Laikipia irrigation field is ___ square metres. I know the formula works for major sectors because ___.	you only the diagram, could you reconstruct the formula from what you know about fractions of a circle?' WAIT 20 SECONDS per group. Before collecting exit prompts, ask one learner to read their second sentence aloud as a class check.		fraction-of-a-circle logic rather than just procedural recall. Collect books or photograph exit prompts to inform the start of Lesson 6.

D. TEACHER REFLECTION

Consider: Did learners instinctively subtract the minor sector area from pi r squared (complement reasoning) or did they apply the formula directly with theta = major angle? Both strategies are valid — highlight both in Lesson 6. Were learners confident naming reflex angles? If not, revisit angle vocabulary at the start of Lesson 6. Check exit prompts for the Laikipia calculation: if many learners used 110 degrees instead of 250, open Lesson 6 with a brief whole-class correction. Finally, note which new DQB questions mention chords or segments — these preview the phenomenon for Lesson 6 and can be used to motivate that lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A large piece of pizza (or circular cut-out) remaining after a minor sector is removed has a central angle greater than 180 degrees, making it a major sector; its angle and the minor sector angle together always equal 360 degrees.
What did I learn?	The formula A = (theta divided by 360) multiplied by pi r squared calculates the area of any sector — minor or major — because theta divided by 360 represents the fraction of the whole circle regardless of the size of the angle; the areas of a major and minor sector of the same circle always sum to pi r squared.
How does this explain the phenomenon?	The unwatered portion of a centre-pivot irrigation field in Laikipia and the return-stroke sweep of a matatu windscreen wiper are both real-world major sectors whose areas can be calculated precisely using the same formula derived from the pizza-slicing phenomenon.

LESSON 6 (80 minutes): Area of a Minor Segment — The Watermelon Rind Problem	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to derive and apply the formula for the area of a minor segment by recognising that a segment is the region between a chord and its arc — equivalent to a sector minus a triangle — and to apply this understanding to real-world Kenyan contexts such as watermelon rind slices.
Knowledge	By the end of the lesson, the learner should be able to: (a) distinguish a minor segment from a sector and a major segment; (b) state that the area of a minor segment = area of sector − area of triangle = $(\theta/360°)\pi r^2 - \frac{1}{2}r^2\sin\theta$; (c) apply the minor segment formula to calculate areas in practical Kenyan contexts.
Skills	By the end of the lesson, the learner should be able to: (a) decompose a circle diagram into a sector and an isosceles triangle; (b) substitute values of r and θ correctly into both the sector and triangle area formulae; (c) subtract to find the minor segment area; (d) interpret calculated segment areas in context.
Attitudes	The learner appreciates how breaking a complex shape into simpler known shapes (sector and triangle) is a powerful mathematical problem-solving strategy; and values the relevance of geometry in everyday Kenyan life, including farming, food preparation, and design.
Key Inquiry Question	How do we calculate the area of a part of a circle? How do areas of sectors and segments of a circle apply in day-to-day activities?
Purpose in Storyline	In Lessons 1–5, learners explored the full circle, then sectors, building up the formula $A = (\theta/360°)\pi r^2$. In Lesson 6, learners now observe that cutting a chord across a sector creates a new region — the minor segment — which is the area between the chord and the arc. Using the watermelon rind as a tangible anchor back to the pizza phenomenon, learners discover they must subtract the isosceles triangle formed by two radii and the chord from the sector area. This advances the driving question: can we predict the exact area of any curved slice, including the rind region, using only r and θ?
Safety Notes	If real watermelon or chapati is brought to class, ensure a clean cutting surface is used and that a teacher or designated adult handles any knife or blade. Remind learners not to eat food used as a learning prop until the teacher gives permission, to maintain a hygienic and focused learning environment.

B. LESSON OVERVIEW
Learners begin by revisiting the pizza/chapati phenomenon and the sector area formula from Lesson 5. The teacher introduces a watermelon cross-section as a new context: when a straight cut (chord) is made across a circular slice of watermelon, the green rind region between the chord and the arc is a minor segment. Through a guided Observe–Explain cycle, learners first predict the area of the rind region, then observe a step-by-step geometric decomposition (sector minus triangle), use the triangle area formula $A = \frac{1}{2}r^2\sin\theta$ alongside the sector formula to calculate the segment area, and finally apply the combined formula to solve contextual problems. The lesson closes with a model revision on the Driving Question Board, connecting the new segment formula to the growing class model of circle-area concepts.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners are shown a large drawn	Display the watermelon diagram	Strategy: Anchored Prediction with	Observable indicator: Can the

📹 **VIDEO:** [Determining the Unknown Angle in an Isosceles Triangle] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Logarithm Triangle (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	diagram on the board (or a printed handout) of a circular watermelon cross-section of radius 14 cm. A horizontal chord is drawn, creating a minor segment (the green rind region at the bottom). Learners are asked: 'Without any formula, estimate the area of the rind segment in cm^2. Write your prediction in your exercise book, draw the diagram, and annotate it with what you know so far.' Learners work individually for 4 minutes, then share predictions with a shoulder partner for 2 minutes. Selected learners share their reasoning with the class. Learners record predictions on a sticky note and place it on the class Prediction Wall under the heading: 'Our best guess for the rind area.'	clearly. Say: 'Look at this watermelon slice. The radius is 14 cm and the central angle of the sector that contains this rind region is 60°. Before we do any maths, I want your best scientific estimate — use what you already know about sectors and triangles.' Allow 10 seconds of silent thinking (WAIT TIME). Cold-call 3 learners using the register — one high-confidence, one mid-confidence, one who has been quieter this week. Ask each: 'What is your estimate, and what did you use to get there?' Record all estimates on the board without evaluating them. Say: 'By the end of this lesson, we will know whose estimate was closest — and more importantly, WHY the formula works.'	Annotated Diagrams. Learners annotate the watermelon diagram with known quantities ($r = 14$ cm, $\theta = 60°$) and shade the region they are estimating. This activates prior knowledge of sectors and triangles simultaneously, surfaces misconceptions (e.g. 'the segment is just a triangle'), and creates cognitive dissonance that motivates the Observe phase.	learner correctly identify the segment as distinct from the full sector? Listen for whether learners distinguish 'the rind part' from 'the whole slice including the red part'. If more than half the class treats segment = sector, pause and draw both regions explicitly before moving on.
Observe Phase 📹 **VIDEO:** [Determining the Unknown Angle in an Isosceles Triangle] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Logarithm Triangle (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Part A — Geometric Decomposition (10 min): Learners receive a printed or drawn worksheet showing three diagrams side by side: (1) a sector OAB with radius r and central angle θ; (2) an isosceles triangle OAB formed by the two radii and the chord AB; (3) the minor segment — the region that remains when the triangle is removed from the sector. Learners colour-code each region: sector in yellow, triangle in orange, segment in green. They write the area formula for each region they already know beneath the diagram. Part B — Formula Derivation (10 min): The teacher guides learners through the derivation on the board step by step. Learners copy and complete a structured fill-in-the-blank derivation in their books: 'Area of minor segment = Area of ______ − Area of ______. = $(\theta/360°) \times \pi \times r^2$	Distribute worksheets. Say: 'Use three different colours to shade the sector, the triangle, and the segment. You have 3 minutes — go.' Set a visible 3-minute timer. After colouring, cold-call 2 learners to name the area formula for the sector — they should say '$(\theta/360°)\pi r^2$' from Lesson 5. Then ask: 'What is the area of triangle OAB if OA = OB = r and the angle at O is θ?' Allow 15 seconds WAIT TIME. Cold-call 2 learners. Guide toward $A = \frac{1}{2}r^2\sin\theta$ if not reached. Say: 'So if I take the sector and I remove the triangle, what am I left with?' Cold-call 1 learner. Then write the full derivation step by step on the board, pausing at each equals sign and asking the class to complete the next line chorally. For the watermelon worked example say: 'Substituting $r = 14$, $\theta = 60°$: what is $(60/360) \times \pi \times 14^2$? Calculate that now.' Allow 2	Strategy: Colour-Coded Decomposition and Structured Fill-in-the-Blank Derivation. Colour-coding makes the geometric subtraction (sector − triangle = segment) visually explicit, reducing cognitive load. The fill-in-the-blank derivation scaffolds algebraic manipulation while ensuring every learner writes the formula in their own book, building procedural fluency alongside conceptual understanding.	Observable indicator: When computing $\frac{1}{2}r^2\sin\theta$, can learners correctly recall that sinθ applies to the included angle at the centre and substitute values accurately? Circulate and check at least 6 books during the 2-minute calculation window. Flag any learner who uses cosθ instead of sinθ — this is the most common error. If observed in 3 or more books, pause the class and re-derive the triangle area formula using the general formula $A = \frac{1}{2}ab \sin C$.

	− ½ × r² × sin θ.' Learners immediately apply the formula to the watermelon example ($r = 14$ cm, $\theta = 60°$) and compare the calculated answer to their earlier prediction.	minutes of silent working. Cold-call 1 learner for the sector area value (~102.6 cm²). Then ask: 'Now calculate ½ × 14² × sin60°. What do you get?' Cold-call 1 learner (~84.87 cm²). Finally: 'So the segment area is approximately…?' Cold-call 1 learner (~17.7 cm²). Write the answer on the board next to the prediction wall estimates.		
Explain Phase 📹 **VIDEO:** Determining the Unknown Angle in an Isosceles Triangle Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Logarithm Triangle (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of 4 to solve two contextual problems that deepen and extend understanding. Problem 1 (Context — Farming): A circular irrigation field in Rift Valley has a radius of 21 m. A straight fence-line (chord) divides off a minor segment for a vegetable garden. The central angle is 90°. Calculate the area of the vegetable garden in m² (give answer to 1 decimal place). Problem 2 (Context — Design/Art): A Maasai beadwork designer in Kajiado creates a circular brooch of radius 5 cm. She cuts off a minor segment with a central angle of 120° to create a flat edge for pinning. Calculate the area of the segment that is removed. After solving both problems, each group writes a one-sentence explanation of why the minor segment area formula always requires subtracting the triangle: 'The segment is smaller than the sector because…' Groups share their sentence with the class.	Assign groups. Display both problems on the board. Say: 'You have 12 minutes to solve both problems as a group. Every member must write the full working in their own book — not just the answer. I will cold-call any member of any group to explain a step.' Circulate actively; stop at each group for 60–90 seconds. Prompt with: 'What is your value of sinθ for this angle?' and 'Have you checked your units?' After 12 minutes, cold-call one member (not the loudest) from two different groups to present full working for one problem each on the board. After each presentation ask the class: 'Do you agree with this answer? Thumbs up, thumbs sideways, or thumbs down — show me now.' Address any disagreements immediately. Then cold-call 2 groups to read their one-sentence explanation. Ask the class: 'Which sentence is clearer? Why?' Converge on a class consensus sentence and write it on the DQB anchor strip.	Strategy: Contextualised Group Problem-Solving with Accountability Cold-Calling. Setting problems in Kenyan farming and design contexts shows learners that the segment formula is not abstract — it is the tool needed whenever a curved boundary meets a straight boundary in real life. Accountability cold-calling (any member can be asked) ensures individual preparation within the group setting, reinforcing both collaborative and individual competency.	Observable indicator: Can every group member write full working (formula stated → values substituted → sector area calculated → triangle area calculated → subtraction performed → answer stated with units) without copying from a peer? During circulation, look for learners who jump to the final answer without showing intermediate steps — prompt them: 'Show me the sector area alone first, then the triangle area.' Collect 3–4 books at random at the end of the group work and scan for correct use of sin90° = 1 and sin120° = √3/2 ≈ 0.866.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Determining the	Each learner individually completes a DQB exit card with three sections: (1) EVIDENCE I HAVE NOW: Write the minor segment area formula in your own words and symbols. (2)	Distribute exit cards. Say: 'This is individual — no talking for 4 minutes. I want to know what YOU now understand, not what your group understands.' After 4 minutes, collect cards and read 3	Strategy: Three-Part DQB Exit Card (Evidence / Question Answered / New Question Raised). This metacognitive structure forces learners to explicitly connect today's evidence	Observable indicator: In the 'EVIDENCE I HAVE NOW' section, does the learner write a formula that correctly shows subtraction of the triangle term? Accept symbolic form $A = (\theta/360°)\pi r^2 - \frac{1}{2}r^2\sin\theta$ or a

Unknown Angle in an Isosceles Triangle Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Logarithm Triangle (PLIX) Source: CK-12 🔍 Search ARES for similar readings	QUESTION ANSWERED: Which part of the driving question can I now answer? Write one full sentence. (3) NEW QUESTION RAISED: What new question does today's lesson make you wonder about? Learners post their exit cards on the class DQB under the column 'Lesson 6 — Segment.' The teacher reads 3–4 cards aloud, highlights new questions raised (e.g. 'What about the major segment?' or 'Can I use this for the pizza's leftover region?'), and pins these as 'Questions to Investigate' for Lesson 7.	selected ones aloud — choose one strong response, one partial response, one that raises an interesting new question. For the partial response, ask the class: 'What would you add to make this more complete?' Cold-call 2 learners. Pin all cards on the DQB. Point to the driving question posted at the top of the board and say: 'Look at how far we have come since Lesson 1 when we could only guess. Now we can calculate the exact area of the rind of a watermelon, the irrigation segment in Rift Valley, and any minor segment — as long as we know r and θ. What part of the pizza question can we now answer that we could not answer in Lesson 1?' Allow 10 seconds WAIT TIME. Cold-call 2 learners.	to the overarching driving question, articulate their current understanding in their own words, and identify the frontier of their knowledge — modelling the authentic scientific practice of using evidence to update and extend understanding.	clear verbal equivalent. Flag any learner who writes only the sector formula with no triangle subtraction — this indicates the core conceptual shift (segment ≠ sector) has not yet been made. Provide these learners with a re-drawn diagram during the next lesson's warm-up.
Model Building Phase 📹 **VIDEO:** Determining the Unknown Angle in an Isosceles Triangle Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Logarithm Triangle (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their cumulative circle-area model that they have been building since Lesson 1 (a large annotated diagram of a circle). They now add a new annotated region: the minor segment, shaded in green. Beside it they write: (a) the formula $A_{segment} = (\theta/360°)\pi r^2 - \frac{1}{2}r^2\sin\theta$; (b) a labelled diagram showing the chord AB, arc AB, centre O, radius r, and central angle θ; (c) a real-life Kenyan example (watermelon rind, irrigation boundary, or beadwork design) in their own words. In the final 3 minutes, pairs compare their updated models and check each other's formula and diagram for accuracy using a peer-check checklist provided by the teacher.	Say: 'Open your circle model from Lesson 1. You are going to add today's discovery to it — the minor segment. You have 10 minutes. Your model must show the labelled diagram, the formula, and a Kenyan real-life example. Be as clear and accurate as possible because you will use this model in your end-of-unit assessment.' Circulate and check that learners are labelling the chord (not calling it a diameter), that θ is marked at the centre O (not at the circumference), and that the formula shows subtraction. After 10 minutes, say: 'Swap your model with your partner. Using the checklist on the board — diagram correctly labelled? Formula correctly written? Real-life example given? — tick or correct their model.' Read the checklist items aloud one at a time. After 3 minutes of peer checking, ask: 'Did	Strategy: Cumulative Visual Model Revision with Peer-Check Checklist. Building a single cumulative circle-area model across all 10 lessons transforms isolated lesson content into a coherent conceptual structure. Peer checking with a specific checklist develops mathematical communication skills and gives learners immediate corrective feedback, reducing the persistence of notation and labelling errors into the assessment.	Observable indicator: Does the learner's model diagram correctly distinguish the chord from the arc, label the centre O and the radii OA and OB, and shade only the segment (not the full sector)? During the peer-check phase, listen for pairs catching the error of θ being placed at the circumference — if this is widespread (3+ pairs), pause the class and re-clarify that the central angle is always at the centre of the circle, consistent with the sector formula from Lesson 5.

		your partner find any errors in your model? What was the most common mistake you saw?' Cold-call 3 learners. Close by saying: 'In Lesson 7, we will investigate what happens to the OTHER region — the major segment — and whether its formula follows the same logic.'		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) During Phase 2, were learners able to recall the triangle area formula $A = ½r^2\sin\theta$ independently, or did they require significant scaffolding? If scaffolding was needed, consider adding a brief warm-up in Lesson 7 that revisits the general triangle area formula $A = ½ab \sin C$ before introducing the major segment. (2) In Phase 3, did learners correctly evaluate $\sin 90° = 1$ and $\sin 120° = \sqrt{3}/2$ without a calculator? Note which trigonometric values caused difficulty — these gaps will compound in Lesson 7 and in Sub-Strand 2.4 Trigonometry. (3) Did the watermelon rind context successfully anchor the abstract formula to a tangible object? If learners still struggled to visualise the segment, consider bringing a physical watermelon or circular cardboard cut-out to Lesson 7 to make the major segment equally concrete. (4) Review the DQB exit cards: how many learners correctly wrote the subtraction formula versus only the sector formula? This ratio determines whether you need a 5-minute consolidation warm-up at the start of Lesson 7. (5) Were group discussions genuinely collaborative or did one member dominate? Consider assigning rotating roles (Scribe, Calculator, Presenter, Checker) in Lesson 7 to distribute participation more equitably across all learners.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed a geometric decomposition diagram showing that a minor segment of a circle (modelled as a watermelon rind slice) is formed by removing an isosceles triangle (with two sides equal to the radius r and included angle θ) from a sector of the same circle.
What did I learn?	Learners derived and applied the formula: Area of minor segment = $(\theta/360°) \times \pi r^2 - ½r^2\sin\theta$, understanding that the sector area gives the 'pie-slice' region and the triangle area ($½r^2\sin\theta$) gives the inner triangular part that must be subtracted to isolate the curved rind region.
How does this explain the phenomenon?	Learners explained why the minor segment is always smaller than its corresponding sector (because a positive triangle area is subtracted), applied the formula to Kenyan farming and design contexts (Rift Valley irrigation field and Maasai beadwork brooch), and updated their cumulative circle-area model to include the annotated minor segment with its formula and a real-life Kenyan example.

LESSON 7 (80 minutes): Area of a Major Segment — Circular Plot Allocation in the Rift Valley	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	By the end of the lesson, learners should be able to calculate the area of a major segment of a circle and apply this knowledge to solve real-world land allocation and design problems.
Knowledge	a) Define a major segment as the region of a circle on the larger side of a chord. b) State that the area of a major segment = area of the full circle − area of the minor segment. c) Recall that area of a minor segment = area of sector − area of triangle.
Skills	a) Calculate the area of a major segment given the radius and central angle. b) Apply the major segment formula to solve problems involving circular plots of land in the Rift Valley. c) Construct and interpret diagrams of major and minor segments accurately.
Attitudes	a) Appreciate the relevance of circle geometry in practical land-use planning and design in Kenya. b) Show patience and precision when carrying out multi-step calculations involving π, sectors, and triangles.
Key Inquiry Question	1. How do we calculate the area of a part of a circle? 2. How do areas of sectors and segments of a circle apply in day-to-day activities?
Purpose in Storyline	In Lesson 7, learners move from calculating minor segments (Lesson 6) to deriving the area of a major segment — the larger portion of a circular region cut by a chord. Using the anchoring phenomenon of the circular pizza/chapati, learners recognise that the larger 'leftover' slice after cutting off a small piece is a major segment. They then transfer this understanding to a Rift Valley land-allocation scenario, where a farmer divides a circular plot using a straight fence (chord), and the larger portion must be calculated for planting maize.
Safety Notes	No hazardous materials are used. Ensure compasses have blunt points for younger learners. Remind students to handle protractors and rulers carefully to avoid injury.

B. LESSON OVERVIEW
Learners revisit the anchoring phenomenon — the circular chapati/pizza cut into unequal slices — and focus on the region that remains after a minor segment is removed. Through a guided problem set framed around circular farming plots in the Rift Valley, learners derive and apply the relationship: Area of Major Segment = πr^2 − Area of Minor Segment, where Area of Minor Segment = Area of Sector − Area of Triangle. The lesson uses collaborative problem-solving, diagram annotation, and a Kenya land-use case study to cement multi-step segment calculations. By the end, learners update their Driving Question Board (DQB) with new evidence about how segment areas help answer questions of proportionality and exact area prediction from the opening phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Proof: radius is perpendicular to a chord it bisects] Source: Khan	Learners are shown a large diagram on the board: a circle of radius 7 m representing a circular shamba (farm plot) in Nairobi's outskirts. A chord divides it into a small region and a large region.	Display the diagram clearly on the board or projector. Say: 'Look carefully at this circular shamba. A straight fence — a chord — separates the sukuma wiki patch from the maize field. In your	Strategy: Anchored Prediction with Peer Justification. Learners activate prior knowledge of sector and minor segment areas (Lessons 4–6) to reason proportionally about the major	Observable indicator: Learners' written predictions show use of proportional reasoning (e.g., 'the chord is close to the edge so most of the circle is maize — maybe about ¾'). Listen for whether

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Circle Graphs to Make Bar Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The small region (minor segment) is shaded green (planted with sukuma wiki). Learners are asked: 'The farmer plants sukuma wiki in the small green region. The rest — the major segment — will be planted with maize. Without calculating, predict: is the maize area more than half the total plot? Is it roughly ¾ of the plot? Write your prediction and your reasoning in your exercise book.' Learners write individually for 3 minutes, then share predictions with a shoulder partner for 2 minutes.	exercise book, write your prediction: how much of the total plot area do you think the maize field takes up? Give a fraction or percentage AND explain your reasoning. You have 3 minutes — go.' [Allow 3 minutes silent writing. Then say:] 'Now share your prediction with your shoulder partner. Convince them with your reasoning.' [Allow 2 minutes pair discussion.] Cold-call 3 learners: 'Amina, what did you predict and why?' — 'Otieno, do you agree or disagree with Amina?' — 'Chebet, what was your reasoning?' [Allow 10–15 seconds WAIT TIME after each question before prompting.] Record two or three contrasting predictions on the board without evaluating them. Say: 'We will return to these predictions at the end of the lesson to see who was closest — and why the maths gives us certainty.'	segment before formal calculation. Connecting the prediction to the shamba context makes the unknown feel concrete and motivating, and surfaces any lingering misconceptions about proportionality from the driving question.	learners spontaneously reference the minor segment or sector formula from previous lessons — this reveals the depth of prior knowledge consolidation. Flag learners who predict exactly ½ without reasoning for targeted support during Phase 3.
Observe Phase 📹 **VIDEO:** Proof: radius is perpendicular to a chord it bisects Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Circle Graphs to Make Bar Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a printed or hand-drawn activity sheet with three diagrams of circles representing circular plots of land in the Rift Valley (Nakuru, Eldoret, and Kericho districts). For each plot, the radius and the central angle subtended by the chord (i.e., the angle of the minor sector) are given. Learners are guided to: (1) Label the minor segment and the major segment on each diagram using colour pencils. (2) Use the provided step-by-step scaffold to calculate the area of the minor segment for Plot A (radius = 7 m, θ = 60°). Scaffold steps on the sheet: Step 1 — Area of full circle = πr². Step 2 — Area of minor sector = (θ/360°) × πr². Step 3 — Area of triangle = ½r²sinθ. Step 4 — Area of minor segment = Step 2	Distribute activity sheets. Say: 'You are now land surveyors working for a county government in the Rift Valley. Three farmers have circular plots and want to know exactly how much land — in square metres — they will have for maize after fencing off a smaller section. Your job is to calculate the major segment area for each plot.' [Point to the scaffold on the sheet.] 'For Plot A, follow every step on your sheet with your partner. I want to see full working — no steps skipped.' [Circulate for 6–8 minutes, checking pair work on Plot A.] Ask targeted questions: 'What formula are you using for the triangle area? Why do we subtract the triangle from the sector?' [After pairs complete Plot A, say:] 'Now attempt Plots B and C on your own	Strategy: Scaffolded Multi-Step Decomposition with Colour Coding. Breaking the major segment calculation into five labelled steps prevents cognitive overload and makes the logical chain visible — learners see that the major segment is built by subtracting, not by a new standalone formula. Colour-coding minor vs. major segments on the diagram makes the geometric relationship tangible and connects to the pizza phenomenon (the large leftover slice is the major segment).	Observable indicator: By the end of Phase 2, learners can independently reproduce all five steps for Plots B and C with correct substitution of values. Specific check: in Step 3, learners must write ½ × r² × sinθ (not ½ × base × height using Pythagoras) — if learners revert to Pythagoras, this signals a gap from Lesson 6 that requires immediate re-teaching. Collect 4–5 exercise books at random to check for correct use of π = 22/7 or 3.14 and proper rounding to 2 decimal places.

	− Step 3. Step 5 — Area of major segment = Step 1 − Step 4. Learners work in pairs to complete Plot A using the scaffold, then attempt Plots B ($r = 10$ m, $\theta = 90°$) and C ($r = 14$ m, $\theta = 120°$) independently, recording all working clearly.	— show all five steps. You have 10 minutes.' [Circulate and identify 2 pairs with correct working and 1 pair with a common error for use in Phase 3.] After independent work, cold-call 2 learners to write their Plot B working on the board step by step. Say: 'Wanjiru, please write Step 1 and Step 2 for Plot B on the board.' [Allow 10 seconds WAIT TIME.] 'Kipchoge, write Steps 3, 4, and 5.' Instruct the class: 'Check your own working against what is on the board. Mark any step where yours differs — we will discuss those differences now.'		
Explain Phase 📹 **VIDEO:** Proof: radius is perpendicular to a chord it bisects Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Circle Graphs to Make Bar Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class discussion to formalise the major segment formula. Learners then apply it to a rich Kenyan context problem: 'A farmer in Eldoret has a circular plot of radius 21 m. She divides it with a straight fence so that the minor sector has a central angle of 80°. She allocates the major segment to growing maize and the minor segment to a vegetable garden (sukuma wiki and tomatoes). Calculate: (a) the area of the major segment, and (b) the difference in area between the two segments.' Learners work through this in groups of four, with each group member assigned a role (Calculator, Diagram Drawer, Step Recorder, Presenter). Groups then compare answers across the class. A class discussion addresses: 'Was your prediction from Phase 1 correct? Is the maize field more than ¾ of the plot? How do you know with certainty now?'	Begin by pointing to the board working from Phase 2. Say: 'Let us now write the general formula together.' Build it line by line with the class, asking: 'What is the first thing we always need?' [WAIT 10 seconds.] Cold-call: 'Njeri — what is Step 1 always?' Guide learners to articulate: Area of Major Segment = $\pi r^2 - [(\theta/360°)\times\pi r^2 - \frac{1}{2}r^2\sin\theta]$. Write this on the board and say: 'Copy this formula into your notes with a box around it.' Then introduce the Eldoret farmer problem. Say: 'In your groups of four, assign roles now — Calculator, Diagram Drawer, Step Recorder, Presenter. You have 12 minutes to solve both parts. I want to see a fully labelled diagram AND all five calculation steps.' [Circulate, prompt with: 'What is the area of the full circle first?' and 'What is sin80° — do you need your calculator or table?'] After 12 minutes, ask two groups to present — one with $r = 21$ m done correctly and one group that made an error in the triangle step. Say to the class: 'Group Mara got a different answer from Group Tana for the	Strategy: Error Analysis + Structured Group Roles (Jigsaw Accountability). Deliberately surfacing a group error and comparing two groups' Step 3 working builds metacognitive awareness — learners must evaluate reasoning, not just get an answer. Assigning roles ensures every learner engages with the calculation at depth. Returning to Phase 1 predictions closes the predict–explain loop and directly addresses the driving question's proportionality puzzle.	Observable indicator: During group work, the Diagram Drawer correctly labels the chord, minor segment, and major segment. The Step Recorder writes all five steps with correct formulas. The Presenter can verbally explain why the area of the major segment is found by subtraction — not a direct formula. Exit check: ask 2–3 learners at random to state the major segment formula from memory before moving to Phase 4.

		triangle area. Both, explain your Step 3.' [WAIT 15 seconds after each explanation.] Guide the class to identify the correct method. Return to Phase 1 predictions on the board: 'Now look at our predictions. Amina said ¾ — let's check: what fraction of the total plot area is the maize field in the Eldoret problem? Use your answer.' Cold-call 3 learners to compute the fraction. Say: 'The maths gives us certainty — that is the power of the formula.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Proof: radius is perpendicular to a chord it bisects] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Circle Graphs to Make Bar Graphs (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners individually write on a sticky note (or in their DQB section of their exercise book) their updated answer to the driving question, specifically addressing: (1) 'Can we now predict the exact area of any slice or segment of a circular pizza/chapati — including the large leftover piece? How?' (2) 'A Rift Valley farmer needs to know how much land is in the major segment of their circular plot — what information do they need and what steps do they follow?' Each learner also writes one new question that has arisen for them — something they are still curious or uncertain about. Questions are posted or read aloud and the teacher updates the class DQB chart.	Say: 'Update your Driving Question Board. Answer two prompts in your book: First — can we now fully predict the exact area of any part of a circle from the pizza phenomenon? Write one clear sentence that answers this. Second — write the steps a Rift Valley farmer would follow. And write one new question you still have.' [Allow 5 minutes silent writing.] Cold-call 4 learners to read their updated driving question answer: 'Omondi, read your sentence.' [WAIT 10 seconds.] 'Fatuma, does your answer add anything new?' After 3–4 responses, ask: 'Who has a new question to add to the board?' Collect 4–5 new questions on sticky notes or write them on the DQB chart. Say: 'These new questions — about irregular shapes, overlapping segments, or composite figures — will guide Lessons 8 and 9. Well done for pushing your thinking further.'	Strategy: Driving Question Board Revisit with Cumulative Evidence Tracking. Updating the DQB at Lesson 7 of 10 allows learners to see how their collective understanding has grown — from predicting proportional slices in Lesson 1 to now computing exact areas of major and minor segments. New questions on the board model scientific/mathematical inquiry as an ongoing process, reinforcing that curiosity is a skill.	Observable indicator: Learners' DQB entries explicitly reference both the radius and the central angle as the two pieces of information needed — matching the driving question's framing. New questions should show intellectual progression (e.g., 'What if the chord doesn't pass through the centre?' or 'Can we find the segment area if we are only given the chord length?') rather than repeating earlier questions — this indicates genuine conceptual growth.
Model Building Phase 📹 **VIDEO:** [Proof: radius is	Learners revise and extend their cumulative circle-area model — a annotated diagram they have built across Lessons 1–7. Today they add: (1) A clearly labelled major	Say: 'Open your cumulative model — the annotated circle diagram you have been building since Lesson 1. Today you are adding the major segment. Draw a new	Strategy: Cumulative Annotated Model Revision. Requiring learners to physically add to and colour-code a single evolving diagram across all 10 lessons	Observable indicator: The model clearly distinguishes all five circle-area concepts (full circle, sector, triangle, minor segment, major segment) with correct formulas

perpendicular to a chord it bisects Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Circle Graphs to Make Bar Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	segment with its formula written beside it. (2) A worked example using the Eldoret farmer's plot (r = 21 m, θ = 80°) showing all five steps. (3) A colour-coded key: full circle (blue), sector (yellow), triangle (red), minor segment (green), major segment (orange). Learners also add an arrow connecting the major segment to the driving question written at the top of their model page: 'How can we use the radius and the central angle to calculate the exact area of any circular slice or region?' The arrow should be annotated with a sentence explaining how today's lesson advances the answer.	circle, label the chord, shade the major and minor segments in your colour-coded key colours, and write the formula for the major segment beside it. Then write your full Eldoret farmer working underneath as the worked example. Finally, draw an arrow from your major segment formula to the driving question at the top and write one sentence explaining what this lesson adds to your answer.' [Allow 8–10 minutes.] Circulate and check: 'Is your formula correct? Is the Eldoret working using all five steps? Does your annotation connect back to the driving question?' [Cold-call 2 learners to share their annotation sentence aloud.] Say: 'By Lesson 10, your model will be a complete reference for calculating any part of a circle — sector, minor segment, or major segment. You are almost there.'	builds schema integration — each new concept is explicitly linked to prior ones rather than stored in isolation. The arrow annotation to the driving question ensures learners continuously evaluate their growing understanding against the central problem, which is the hallmark of inquiry-based sensemaking.	and colour codes. The annotation arrow contains a meaningful sentence — e.g., 'The major segment formula completes our toolkit because now we can find the area of the larger OR smaller part of any circle cut by a chord, not just a pizza slice cut from the centre.' Collect 5 models at the end of class to assess completeness and conceptual accuracy before Lesson 8.

D. TEACHER REFLECTION
After the lesson, reflect on the following: (1) Were learners able to move fluently between the five steps without returning to the scaffold by Plot C? If not, which step — typically Step 3 ($\frac{1}{2}r^2\sin\theta$) — caused the most confusion, and how will you address it at the start of Lesson 8? (2) Did the group role structure (Calculator, Diagram Drawer, Step Recorder, Presenter) ensure equitable participation, or did one or two learners dominate? Consider reassigning roles more deliberately in Lesson 8. (3) When learners updated the DQB, did their new questions reveal genuine conceptual progression, or were they repeating earlier uncertainties? Use new questions to calibrate the complexity of Lesson 8's problems. (4) Did the Rift Valley land-allocation context feel authentic and engaging to learners, or did it feel contrived? Gather learner feedback informally at the door — 'Give me one thumb up if the farming context helped you understand the maths.' (5) Were predictions from Phase 1 revisited meaningfully in Phase 3, or did the connection feel rushed? If rushed, build more explicit prediction-revisitation time into the Phase 3 plan for Lesson 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed a circular shamba (farm plot) divided by a chord into a small sukuma wiki patch and a large maize field, and worked through three Rift Valley circular plot diagrams (Nakuru, Eldoret, Kericho) to gather evidence about how the major segment relates to the full circle and minor segment.
What did I learn?	Learners learned that the area of a major segment = πr^2 − (area of minor segment), where area of minor segment = $(\theta/360°)\times\pi r^2 - \frac{1}{2}r^2\sin\theta$. They applied this five-step process to calculate exact land areas for Rift Valley farming plots and confirmed that the larger

	region of a circle cut by a chord can be found precisely using only the radius and the central angle.
How does this explain the phenomenon?	Learners explained that any circular region divided by a chord produces a minor and a major segment, and that the major segment area is determined by subtracting the minor segment from the total circle area. They connected this to the driving question by demonstrating that — just as with pizza slices — knowing only the radius and central angle is sufficient to calculate the exact area of either segment, enabling precise land allocation, design, and resource planning in real Kenyan contexts.

LESSON 8 (80 minutes): Application & Model Building — Solving Real-Life Problems Using Arc Length, Sector Area, and Segment Area

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	By the end of this lesson, the learner should be able to apply the formulas for arc length, sector area, and segment area to solve multi-step real-life problems, including tiling a circular garden, pricing watermelon slices fairly, and designing a Maasai bead pattern.
Knowledge	Learners will consolidate knowledge of: (a) sector area formula $A = (\theta/360°) \times \pi r^2$; (b) arc length formula $L = (\theta/360°) \times 2\pi r$; (c) segment area = sector area − area of triangle ($A_{segment} = (\theta/360°) \times \pi r^2 - \frac{1}{2}r^2\sin\theta$); and (d) the proportional relationship between central angle and area of a sector.
Skills	Learners will: (a) identify given information (r and θ) from a real-life problem context; (b) select and apply the correct formula for arc length, sector area, or segment area; (c) execute multi-step calculations accurately; (d) interpret and communicate answers in context; (e) revise their cumulative model on the Driving Question Board.
Attitudes	Learners will: (a) appreciate the relevance of circle geometry in Kenyan daily life — farming, food pricing, and cultural design; (b) show confidence and persistence in tackling multi-step problems; (c) collaborate respectfully, valuing each group member's mathematical contribution.
Key Inquiry Question	How do areas of sectors and segments of a circle apply in day-to-day activities?
Purpose in Storyline	Lesson 8 is the APPLICATION & MODEL BUILDING lesson in the 10-lesson evidence-gathering sequence. Students have previously derived the sector area formula ($A = (\theta/360°) \times \pi r^2$), confirmed proportionality, calculated arc lengths, and found segment areas. Today they synthesise ALL formulas to solve three authentic Kenyan multi-step problems — tiling a circular garden in the Rift Valley, pricing watermelon slices at a Nairobi market stall, and designing a Maasai bead pattern. By the end of the lesson learners revise their DQB models, bringing the class measurably closer to a complete answer to the driving question.
Safety Notes	No hazardous materials are used in this lesson. If physical watermelon or chapati is brought in for motivation, ensure learners with food allergies are identified in advance. Compass and protractor points are sharp — remind learners to place instruments on the desk when not in use and never to point them at others.

B. LESSON OVERVIEW

Lesson 8 is the synthesis and application lesson in the Area of Part of a Circle sequence. Students open by revisiting the pizza/chapati phenomenon and the class Driving Question Board, then make a brief prediction about three real-life Kenyan scenarios. They move through a structured group-work Observe phase in which each group solves one of three contextualised multi-step problems (circular garden tiling in the Rift Valley, fair pricing of watermelon slices at a Nairobi market, Maasai bead-pattern design), applying arc length, sector area, and segment area formulas. In the Explain phase, groups present solutions, the class critiques working, and the teacher leads whole-class sensemaking using the 'Formula Selection Chart' strategy. The DQB Creation phase involves groups adding new evidence sticky notes and revising the class model. The Model Building phase closes the lesson: learners individually update their personal cumulative diagrams and write a one-sentence answer to the driving question, which is shared and displayed. Throughout, the anchoring phenomenon (unequal pizza/chapati slices) is repeatedly referenced to ground all abstract formulas in a concrete, proportional reality.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Video: Balancing Act Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Quadratic Formula (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students enter to find the original circular chapati (or a drawn representation on the board) displayed at the front, now divided into five unequal slices with central angles labelled: 40°, 55°, 70°, 80°, and 115°. The radius is marked as 15 cm. On their individual mini-whiteboards (or a half-sheet of paper), learners respond to THREE quick prediction prompts written on the board: (1) GARDEN: A circular shamba (farm plot) in Nakuru has radius 7 m. A farmer wants to tile ONLY a sector of angle 120° with paving stones. Predict — will the paving cover more or less than one-third of the full garden floor? Write your reasoning in one sentence. (2) MARKET: A watermelon vendor in Gikomba Market cuts a circular watermelon (radius 20 cm) into two slices — one with central angle 80° and one with 160°. Predict — is the bigger slice exactly twice the area of the smaller slice? Yes / No / Depends — explain why. (3) BEAD DESIGN: A Maasai beadwork teacher in Kajiado wants to fill a segment (NOT the full sector) of a circular bead frame with red beads. The radius is 10 cm and the central angle is 60°. Predict — is the segment area less than, equal to, or greater than half the sector area?	SETUP (before class): Draw or display the chapati diagram prominently. Write the three prediction prompts clearly on the board with diagrams. OPENING (1 min): Say exactly — 'Last week we cut this chapati and asked: can we predict the exact area of any slice? Today we find out if we can — not just for food, but for a farm in Nakuru, a market in Gikomba, and a bead design in Kajiado. Before we calculate anything, I want your predictions.' SILENT PREDICTION (3 min): Circulate silently. Do NOT confirm or correct. Note 2–3 interesting (especially incorrect) predictions to return to later. PAIR SHARE (2 min): 'Turn to your partner. Share your prediction for Question 2 — the watermelon. Do you agree? You have 90 seconds.' CLASS TEMPERATURE CHECK (2 min): 'Hands up — who said the bigger watermelon slice is EXACTLY twice the area? ... Who said it DEPENDS? ... Who said YES, always?' [Count hands aloud — e.g., '14 say exactly twice, 8 say depends.']. Do NOT reveal the answer. Say: 'Hold that thought — your evidence today will tell you who is right.' BRIDGE (1 min): 'By the end of	Strategy: PREDICTION TEMPERATURE CHECK — Students publicly commit to a prediction before instruction. This activates prior knowledge of proportionality and the sector formula, creates cognitive dissonance (especially on the watermelon question, where the answer IS exactly twice — confirming linearity), and gives the teacher real-time diagnostic information about residual misconceptions. The public 'hands up' count makes predictions social and memorable, increasing motivation to verify them during the Observe phase.	Observable indicator: During circulation, check that at least 80% of learners can articulate a reason (even if incorrect) for their prediction — not just a guess. Specifically listen for whether learners spontaneously invoke $A = (\theta/360°) \times \pi r^2$ in their reasoning (shows consolidation from Lessons 5–7). RED FLAG: Any learner who says 'area depends on something other than r and θ' needs targeted support during group work.

	Learners write predictions silently for 3 minutes, then share with a seat partner for 2 minutes. Teacher takes a class 'temperature check' by asking for a show of hands on each prediction.	this lesson, each of your groups will have solved one of these three problems completely. Then we share. Let us begin.'		
Observe Phase 📹 **VIDEO:** [Video: Balancing Act] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Quadratic Formula (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students work in pre-assigned groups of 4. Each group receives ONE of three differentiated problem cards (colour-coded: green = Garden, yellow = Market, red = Bead Design). Each card contains a full problem with a diagram, all given information, a 'formula bank' reminder, and three scaffolded sub-questions. Groups must show ALL working. 🟢 GREEN — GARDEN TILING (Rift Valley context): A circular shamba in Nakuru has radius 7 m. A farmer, Mama Wanjiku, wants to tile a sector of central angle 120° with square paving stones, each covering 0.25 m^2. (a) Calculate the area of the sector. [Use $A = (\theta/360°) \times \pi r^2$] (b) How many paving stones does Mama Wanjiku need? (Round up to the nearest whole stone.) (c) The arc of the sector will be edged with a decorative border. Calculate the arc length. 🟡 YELLOW — WATERMELON PRICING (Nairobi market context): A circular watermelon at Gikomba Market has radius 20 cm. It is cut into two unequal slices: Slice A has central angle 80°, Slice B has central angle 160°. The vendor charges Ksh 5 per cm^2 of cross-sectional area.	DISTRIBUTION (1 min): Hand out colour-coded problem cards and A3 paper. Say: 'Each group has a different real Kenyan problem. In 18 minutes, your group must solve all three sub-questions AND be ready to present your solution to the class in two minutes. I will be visiting every group.' CIRCULATION PROTOCOL — visit each group at least twice using this sequence: — First visit (minutes 1–8): Check that the group has correctly identified r and θ from the problem. Ask: 'What is your radius? What is your central angle? Which formula are you using first?' WAIT 10 seconds for response before prompting. — Second visit (minutes 9–16): Check arithmetic and units. For GREEN group, prompt: 'Your area is in m^2 — what unit will your number of stones be in?' For YELLOW group: 'You said Slice B is exactly twice Slice A — can you explain WHY using the formula?' For RED group: 'When you subtract the triangle from the sector, what does the remaining area represent physically?' SPECIFIC QUESTIONS TO ASK (verbatim): — 'Show me where the formula $A = (\theta/360°) \times \pi r^2$ comes from — connect it back to our chapati.'	Strategy: DIFFERENTIATED CONTEXTUAL PROBLEM CARDS — Each group tackles a different real-life Kenyan application, ensuring that when groups share in Phase 3, every student encounters all three contexts (garden, market, bead design) and sees the same formulas applied in different ways. The colour-coded scaffolding (sub-questions a, b, c) guides learners from formula recall → single calculation → interpretation/application, mirroring the NGSS practice of 'Using Mathematics and Computational Thinking' while embedding the formulas in authentic Kenyan economic and cultural contexts. The requirement to write working on A3 paper externalises thinking, making it visible for peer and teacher critique.	Observable indicators: — GREEN group: Correct sector area ≈ 51.31 m^2; correct number of stones = 206 (rounded up from 205.24); correct arc length ≈ 14.66 m. FLAG if group divides by 360 without multiplying by πr^2. — YELLOW group: Slice A area ≈ 222.53 cm^2; Slice B area ≈ 445.06 cm^2; Slice B IS exactly twice Slice A (confirming the proportionality prediction). FLAG if group says 'it depends on the radius' — this reveals a residual misconception about proportionality. — RED group: Sector area ≈ 52.36 cm^2; triangle area = 43.30 cm^2 (½ × 100 × sin60°); segment area ≈ 9.06 cm^2. FLAG if group subtracts arc length instead of triangle area — a common procedural error.

	(a) Calculate the area of Slice A. (b) Calculate the area of Slice B. (c) What is the fair price of each slice? Is Slice B exactly twice the price of Slice A? Show your working and explain. 🔴 RED — MAASAI BEAD DESIGN (Kajiado context): A Maasai beadwork teacher in Kajiado is designing a circular bead frame of radius 10 cm. She wants to fill a SEGMENT (the region between a chord and its arc) with red beads. The chord subtends a central angle of 60° at the centre. (a) Calculate the area of the sector with angle 60°. (b) Calculate the area of the triangle formed by the two radii and the chord. [Use $A_{triangle} = \frac{1}{2}r^2\sin\theta$] (c) Calculate the area of the segment. State what formula you used. All groups record their solutions on large A3 paper to be displayed during the Explain phase. Groups are given 18 minutes of working time, followed by 4 minutes to prepare a 2-minute group presentation.	— 'If the angle doubled but the radius stayed the same, what would happen to the area? How do you know?' — 'Is your answer reasonable? Estimate first — is a 120° sector roughly one-third of the full circle?' COLD CALLS during circulation: Tap individual students (not just group spokespersons) and ask them to explain one step. Aim for at least 3 cold calls per group visit. PRESENTATION PREP (4 min): 'Stop working. You have 4 minutes to agree on who presents which sub-question. Each member of your group must speak for at least 20 seconds.'		
Explain Phase 📹 **VIDEO:** Video: Balancing Act Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos	Each group presents their A3 solution to the class (2 minutes per group = 6 minutes total presentations). After each presentation, the class has 1 minute for peer questions or challenges — the presenting group must defend their working. After all three presentations, the teacher leads a whole-class 'Formula Selection Chart' activity (8	PRESENTATION MANAGEMENT (6 min): Introduce each group — 'Green group — Mama Wanjiku's garden in Nakuru. You have 2 minutes. Begin.' Use a visible timer. After each presentation, immediately open the floor: 'One question or challenge for the Green group — hands up.' Cold-call ONE student who has NOT yet spoken in the lesson. WAIT 10	Strategy: FORMULA SELECTION CHART — This is a 'knowledge organiser built live with students' strategy. Rather than giving learners a pre-made formula sheet, the chart is co-constructed from their own problem-solving evidence. This deepens retention because learners see each formula as an answer to a specific question ('What do I need to find?')	Observable indicators on exit slip: (1) Correct formula written with all symbols defined; (2) Numerical answer correct with correct units (m^2, cm^2, or m/cm); (3) Sentence demonstrates causal reasoning — e.g., 'This connects to our chapati because the sector of the watermelon is the same fraction (80/360) of the full circle as that slice of chapati is of the whole.'

📖 **HTML:** [Quadratic Formula (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	minutes): on the board, a three-column chart is drawn with headers 'WHAT I NEED TO FIND | FORMULA I USE | KENYAN EXAMPLE FROM TODAY.' Students call out entries from the three group problems to populate the chart together. The teacher then poses two bridging questions that connect all three problems back to the anchoring phenomenon: 'Every formula we used today — where did it come from? Let us trace it back to the chapati.' Learners then individually complete a 'Three-Problem Summary' exit slip (4 minutes): write the formula used, the answer, and one sentence connecting the result to the chapati phenomenon for ONE of the three problems.	seconds. FORMULA SELECTION CHART (8 min): Draw the three-column chart on the board. Say: 'Let us organise what we know. Call out: what did the Green group NEED to find?' [Students: 'Sector area.'] 'Which formula?' [Students: 'A = $(\theta/360°) \times \pi r^2$.'] Fill in chart. Repeat for arc length, then for segment area from the Red group. Once chart is complete, ask: 'Look at the Segment Area row. The formula says segment = sector − triangle. Why do we subtract the triangle? What does the triangle represent physically in the bead design?' WAIT 15 seconds. Cold-call 2 students. Expected answer: 'The triangle is the part of the sector that is NOT between the chord and the arc — it is the middle solid part.' BRIDGE TO PHENOMENON (2 min): Point to the chapati diagram. Say verbatim: 'Every formula on this chart started with one observation about our chapati: a sector is a FRACTION of a full circle. That fraction is $\theta/360°$. Everything else we built from there. The garden, the watermelon, the bead design — all of them are just that same fraction, applied to a new context.' Pause 10 seconds. EXIT SLIP (4 min): 'Individually — on your half-sheet, choose ONE of the three problems. Write: (1) the formula you used, (2) the numerical answer with units, (3) one sentence starting with: "This connects to our chapati	rather than an isolated rule to memorise. The Kenyan-example column anchors each formula in a concrete context, satisfying NGSS Practice 6 (Constructing Explanations) and connecting to the KICD competency of 'Critical Thinking and Problem Solving.'	RED FLAG: Any exit slip that states the formula without units or writes '$\theta/360 \times r^2$' without π — indicates the learner has memorised a corrupted version of the formula.

		because..." You have 4 minutes. No group discussion.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Balancing Act] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Quadratic Formula (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students return to the class Driving Question Board (DQB), which has been displayed since Lesson 1 and contains sticky notes from all previous lessons. In groups, learners are given three sticky notes each in two colours: BLUE = 'New evidence we now have' and ORANGE = 'Questions we still have.' Groups spend 5 minutes writing their sticky notes based on today's three problems and adding them to the DQB. The teacher then leads a 5-minute whole-class DQB gallery walk — students read the new additions silently for 2 minutes, then the class votes (thumbs up/down/sideways) on whether today's lesson has FULLY answered the driving question: 'How can we use the radius and the central angle of a circular pizza slice to calculate its exact area — and apply the same method to solve real problems in farming, design, and everyday life in Kenya?' The teacher records the vote count and leads a 2-minute discussion on what (if anything) remains unanswered.	DQB STICKY NOTE WRITING (5 min): Distribute blue and orange sticky notes. Say: 'You need at least ONE blue note — new evidence from today. And at least ONE orange note — a question you still have, or a new question today raised. Think about the garden, the watermelon, the bead design. Go.' Circulate and prompt reluctant writers: 'What surprised you today? What would happen if the sector crossed the centre — could you still use the same formula?' EXAMPLE PROMPTS to push thinking toward orange notes: 'What if the shape was not a sector but an annulus — a ring shape? Could you still use our formula?' 'What if the central angle was bigger than 180°? Is that still a sector?' GALLERY WALK & VOTE (5 min): 'Two minutes — silent read of the new DQB additions. Then I want a vote: thumbs up = yes, today's lesson FULLY answers the driving question; thumbs sideways = mostly but not completely; thumbs down = not yet.' Count and record. Say: 'Most of you said [result]. What is still missing from our answer?' Cold-call 2 students. Expected: 'We have not yet connected segment area to a real design problem fully' OR 'We need Lesson 9 to practise more.' Affirm: 'Exactly — that is what Lessons 9 and 10 are for.'	Strategy: TWO-COLOUR STICKY-NOTE DQB UPDATE — Using two distinct colours for 'evidence' (blue) and 'new questions' (orange) trains learners to distinguish between what they know and what they do not yet know — a metacognitive habit central to scientific inquiry (NGSS Practice 1: Asking Questions). The public vote on whether the driving question is fully answered creates accountability and honest self-assessment, and the teacher's live recording of the vote count models how scientists track the state of their understanding over time.	Observable indicators: (1) Every group posts at least one blue note that correctly references a formula or result from today (e.g., 'We proved the watermelon slice with double the angle has exactly double the area'). (2) Orange notes reveal the depth of curiosity — notes asking about reflex angles, annuli, or 3D applications indicate high-level engagement. (3) The vote outcome — if more than 30% vote 'thumbs down,' the teacher should address the specific gap before Lesson 9. RED FLAG: Groups posting only generic notes like 'we learned about sectors' without referencing a specific calculation — these groups may not have internalised the formula.

		BRIDGE (1 min): Point to the driving question. Say: 'Look how far our DQB has come since Lesson 1. On Day 1 we could not even calculate the area of that small chapati slice. Today you calculated the area of a farm, priced a watermelon, and designed a bead pattern. That is the power of one formula.'		
Model Building Phase 📹 **VIDEO:** Video: Balancing Act Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Quadratic Formula (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner opens their personal 'Cumulative Circle Geometry Model' — a diagram they have been building since Lesson 1 in their exercise books. The model currently shows: a labelled circle with radius r, a sector with angle θ, the sector area formula, the arc length formula, and (from Lesson 7) the segment construction. Today learners add THREE new elements to their personal model: (1) A REAL-WORLD ANNOTATION LAYER: Next to each formula on their model, learners write a one-line Kenyan context where that formula applies — e.g., next to $A = (\theta/360°) \times \pi r^2$, write 'Used to tile Mama Wanjiku's shamba in Nakuru (120°, r = 7 m).' (2) A PROPORTIONALITY STATEMENT: Learners write and box the conclusion: 'If central angle doubles (and r stays the same), sector area EXACTLY doubles — confirmed by the Gikomba watermelon problem.' (3) A ONE-SENTENCE DRIVING QUESTION ANSWER: At the bottom of their model, learners write their current best answer to the driving question — 'The exact	MODEL REVISION INTRODUCTION (1 min): Say verbatim: 'Open your cumulative model. You have been building this since Lesson 1 — every lesson, you have added something. Today you are adding the most important layer: the real world. Three additions — I will put the instructions on the board.' Write on board: '1. Real-world annotation — add a Kenyan example next to EACH formula. 2. Proportionality statement — write and box it. 3. Driving question answer — one sentence, use the formula, mention Kenya.' INDIVIDUAL WORK (10 min): Circulate and read over shoulders. Look specifically for: — Correct formula written with ALL symbols (θ, r, π, 360) — Proportionality statement that says 'exactly doubles' (not 'gets bigger') — Driving question answer that uses the formula AND names a Kenyan context Prompt struggling learners: 'Look	Strategy: CUMULATIVE MODEL REVISION WITH REAL-WORLD ANNOTATION LAYER — The cumulative personal model is the student's growing scientific explanation, revised lesson by lesson (mirroring NGSS Practice 6: Constructing Explanations, and Practice 2: Developing and Using Models). By requiring a 'real-world annotation layer' today, learners are forced to connect every abstract formula to a concrete Kenyan application — preventing the common failure mode of students who can recite a formula but cannot deploy it. The paired comparison of driving question answers introduces peer argumentation (NGSS Practice 7: Engaging in Argument from Evidence), improving precision of mathematical language.	Observable indicators: (1) Personal model contains all three new additions (annotation, proportionality statement, driving question answer). (2) Driving question answer is mathematically complete — includes the formula with symbols, a numerical example or context, and the word 'exactly' or 'proportional' to convey precision. (3) Proportionality statement correctly identifies that the relationship is LINEAR (area doubles when angle doubles, for fixed r). RED FLAG: Any learner who writes 'area depends on the size of the circle' without specifying that r is fixed — return to the Gikomba watermelon problem and ask: 'In that problem, the radius was the same for both slices — what was the ONLY thing that changed?'

	area of any circular slice = (θ/360°) × πr², where θ is the central angle and r is the radius. The same formula applies to farms, food pricing, and bead design in Kenya.' After 10 minutes of individual model revision, learners pair up for 3 minutes and compare their one-sentence driving question answers — improving wording together. Three pairs are cold-called to read their answers aloud. The lesson closes with the teacher reading the class's best collective answer and posting it on the DQB as 'Our Best Answer — Lesson 8.'	at your exit slip from Phase 3 — you already wrote the sentence. Expand it slightly and put it here.' PAIR COMPARISON (3 min): 'Turn to your partner. Read each other's driving question answer. Improve ONE word or phrase together. You have 90 seconds each.' COLD CALLS (2 min): 'Let me hear three driving question answers. [Name 1] — read yours.' WAIT 5 seconds after each for class reaction. 'Does anyone want to suggest an improvement?' After three readings, say: 'I am going to combine the best elements and write our class answer on the DQB. Here it is: [read composite answer].' CLOSING (1 min): 'From a chapati on a table in Lesson 1 to a farm in Nakuru, a market in Gikomba, and a bead design in Kajiado — you have answered the driving question. Next lesson, we will practise and go even deeper. Well done.'		

D. TEACHER REFLECTION

After this lesson, reflect on the following questions before planning Lesson 9:

1. PROPORTIONALITY CONSOLIDATION: During the Yellow (Watermelon) group presentation, did learners spontaneously use the word 'proportional' or 'linear relationship' to explain why Slice B is exactly twice Slice A? If fewer than half the class could articulate this verbally, plan a 5-minute re-teaching opener for Lesson 9 using a simple table: angle 40°, 80°, 120°, 160° → corresponding areas — ask learners to identify the pattern.

2. SEGMENT FORMULA ERRORS: Did the Red (Bead Design) group correctly subtract the triangle area using $\frac{1}{2}r^2\sin\theta$, or did any group attempt to subtract arc length (a dimensional error — cm vs cm²)? If this error appeared, it signals a conceptual gap between 'area' and 'length' that must be explicitly addressed in Lesson 9 with a dimensional analysis activity.

3. DQB QUALITY: Were the orange 'question' sticky notes substantive (e.g., asking about reflex sectors, annuli, or 3D applications) or superficial (e.g., 'I don't understand')? Substantive questions should be incorporated into Lesson 9's opening. Superficial questions indicate learners are not yet comfortable with metacognitive self-assessment — plan a think-aloud modelling of 'how to ask a good mathematical question.'

4. DRIVING QUESTION ANSWER QUALITY: Read all individual 'one-sentence answers' from the Model Building phase. Tally: (a) how many include the formula $A = (\theta/360°) \times \pi r^2$ with all symbols correct; (b) how many mention a Kenyan context; (c) how many include the word 'exactly' or 'precisely.' Learners missing two or more of these criteria should receive targeted feedback written in their books before Lesson 9.

5. TIMING: Was 18 minutes sufficient for group problem-solving in Phase 2? If more than two groups did not reach sub-question (c), consider providing a partial worked example for sub-question (a) in Lesson 9's practice problems to ensure all learners reach the interpretation step.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Three real-life Kenyan problems were solved using circle geometry formulas: (1) Mama Wanjiku's circular shamba in Nakuru — a 120° sector of radius 7 m required approximately 206 paving stones (sector area ≈ 51.31 m^2) and an arc-length border of ≈ 14.66 m; (2) Two watermelon slices at Gikomba Market (radii both 20 cm, angles 80° and 160°) — Slice A area ≈ 222.53 cm^2 priced at Ksh 1,112.65 and Slice B area ≈ 445.06 cm^2 priced at Ksh 2,225.30, confirming Slice B costs exactly twice Slice A; (3) A Maasai bead segment in Kajiado ($r = 10$ cm, $\theta = 60°$) — sector area ≈ 52.36 cm^2, triangle area = 43.30 cm^2, segment area ≈ 9.06 cm^2.
What did I learn?	All three problems were solved using the same two foundational formulas derived from the chapati phenomenon: Sector Area = $(\theta/360°) \times \pi r^2$ and Segment Area = Sector Area − $\frac{1}{2}r^2\sin\theta$. The watermelon problem provided direct, numerical confirmation that the sector area formula is LINEAR in θ for a fixed radius — doubling the central angle exactly doubles the area. This validates the proportionality conjecture first raised in Lesson 2 when students noticed that the wider chapati slices appeared to be 'more than just a bit' larger. The segment formula extends the sector formula by subtracting the triangular region formed by the two radii and the chord, isolating only the curved 'bite' between the arc and the chord.
How does this explain the phenomenon?	The sector area formula $A = (\theta/360°) \times \pi r^2$ works because a sector is simply a fraction (θ/360°) of the full circle's area (πr^2). The fraction θ/360° captures exactly how much of the full rotation the sector occupies — which is why doubling θ exactly doubles the area (for fixed r). The segment formula subtracts the area of the triangle ($\frac{1}{2}r^2\sin\theta$) from the sector area because the triangle occupies the 'filled-in' central portion of the sector that is NOT between the chord and the arc. All three Kenyan problems — the farm, the market, the bead design — are structurally identical to the original chapati phenomenon: a circular region with a known radius is divided by a central angle, and the task is to find the exact area of that portion.

LESSON 9 (80 minutes): Digital Tools & Consolidation: Confirming Sector and Segment Areas, Correcting Errors	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To use digital tools and calculators to verify sector and segment area calculations, identify and correct common computational errors, and consolidate understanding of the sector and segment area formulas in preparation for summative application.
Knowledge	By the end of the lesson, the learner should be able to: (a) apply the formula $A = (\theta/360°) \times \pi r^2$ to calculate the area of a sector using a calculator; (b) apply the area of a sector and the area of a triangle to determine the area of a segment; (c) identify and explain common errors in sector and segment calculations (e.g., using degrees as radians, omitting triangle subtraction).
Skills	By the end of the lesson, the learner should be able to: (a) use a scientific calculator or digital simulation to confirm sector and segment area values; (b) detect and correct errors in worked examples; (c) communicate mathematical reasoning clearly when explaining corrections to peers.
Attitudes	By the end of the lesson, the learner should be able to: (a) appreciate the value of digital tools in verifying mathematical accuracy; (b) demonstrate intellectual honesty by acknowledging and correcting their own prior errors; (c) show responsibility in the careful use of calculators and digital devices.
Key Inquiry Question	1. How do we calculate the area of a part of a circle? 2. How do areas of sectors and segments of a circle apply in day-to-day activities?
Purpose in Storyline	In Lesson 9, learners reach the DIGITAL TOOLS & CONSOLIDATION phase of the sequence. Having hand-calculated sector and segment areas in previous lessons using the pizza/chapati phenomenon, students now use calculators and digital simulations to verify those results, surface and fix systematic errors (degrees vs. radians; forgetting to subtract the triangle from the sector), and update the Driving Question Board with consolidated, confirmed understanding. This lesson bridges the hand-calculation fluency of earlier lessons to confident, accurate problem-solving ready for the summative Lesson 10.
Safety Notes	No physical hazards. If using school devices (tablets, laptops, or phones), remind learners to handle equipment responsibly, keep devices on desks, and avoid accessing content unrelated to the lesson. Ensure calculators are returned to storage after use.

B. LESSON OVERVIEW
Learners begin by revisiting their pizza-slice predictions and hand-calculated values from earlier lessons. They then use scientific calculators and a digital simulation (GeoGebra or equivalent offline tool on ARES) to re-compute sector and segment areas, comparing digital results with their prior hand-calculations. A structured 'Error Autopsy' activity presents four deliberately flawed worked examples — two sector problems and two segment problems — for learners to diagnose and correct in pairs. Common errors addressed include: (1) substituting the angle in degrees directly into a radian-based formula, (2) using diameter instead of radius, (3) forgetting to subtract the triangle area when finding a segment, and (4) rounding π prematurely. The lesson closes with an updated Driving Question Board (DQB) entry and a reflective exit ticket connecting digital confirmation back to the chapati/pizza phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners retrieve their hand-	1. Display the three anonymous	Strategy: ANTICIPATION +	Observable indicator: Learners'

📹 **VIDEO:** [Radians as ratio of arc length to radius] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Quadratic Formula (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	written sector and segment calculations from Lessons 6–8 (pizza/chapati slices with recorded radii and central angles). Each learner silently reviews their own work and writes a one-sentence prediction on a sticky note: 'I think my calculator answer will / will not match my hand calculation because…'. Learners then share predictions with a shoulder partner before a whole-class share-out. The teacher posts three anonymous samples of hand-calculations (from prior lessons) on the board — one correct, one with a degrees-as-radians error, one missing the triangle subtraction — and asks: 'Before we use any tool, which of these three do you trust most, and why?'	worked examples on the board (prepared in advance on large manila paper or projected via ARES). Say: 'Take 2 minutes — no talking — to read all three solutions. Mark any step that looks suspicious with a question mark.' (Allow 2 min silent reading.) 2. Cold-call 3 learners: 'Amina, which solution do you trust most right now, and point to the exact step that makes you trust it.' Repeat for 2 more learners. WAIT TIME: 12 seconds after each question before accepting a response. 3. Do NOT confirm or reject any answer yet. Say: 'Hold your judgment. By the end of today you will be able to prove which solution is correct and explain exactly where the others went wrong.' 4. Distribute sticky notes. Say: 'Write your prediction: will your own earlier answer match what a calculator gives you? One sentence, your reasoning.' Allow 3 minutes. 5. Ask 4 learners to read their sticky notes aloud. Post all sticky notes on the DQB in a 'Predictions — Lesson 9' cluster.	ERROR DETECTION. By asking learners to evaluate flawed examples before using any digital tool, the teacher activates prior knowledge and creates cognitive dissonance — learners must commit to a judgment they cannot yet fully verify. This productive tension motivates careful use of the calculator in Phase 2. Connecting to the phenomenon: the three anonymous solutions are framed as 'three different classmates tried to calculate the area of the wide pizza slice (θ = 120°, r = 14 cm) — let's find out who was right.'	sticky-note predictions reveal whether they can articulate a specific mathematical reason for a potential discrepancy (e.g., 'I think my answer might be wrong because I rounded π to 3.14 early'). Teacher listens for vague vs. specific reasoning during cold-calls. Flag any learner who cannot identify any suspicious step in the three examples — provide a prompt card: 'Check: did each solution use radius or diameter? Did the segment solution subtract something?'
Observe Phase 📹 **VIDEO:** [Radians as ratio of arc length to radius] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Quadratic Formula (Flexbook)] Source: CK-12 🔍 Search ARES	Learners work in pairs. Each pair receives a Verification Worksheet listing four pizza/chapati scenarios (matching scenarios from Lessons 6–8): (A) Sector: chapati radius = 21 cm, θ = 90°; (B) Sector: pizza radius = 14 cm, θ = 120°; (C) Segment: maize storage circle r = 10 m, θ = 60°; (D) Segment: irrigation arc r = 35 m, θ = 150°. Step 1 — Calculator run: Learners use scientific calculators to compute each area step-by-step, recording every intermediate value (θ/360, πr^2, product, triangle area, final segment area). Step 2 — Digital simulation: Using the ARES	1. Before pairs begin, model ONE complete calculator sequence live for scenario (A): Say: 'Watch my fingers on the calculator. I press: 90 ÷ 360 = … I get 0.25. I then press × π × 21 × 21 =. What do I get? Call it out.' Cold-call 2 learners to confirm the display. WAIT TIME: 10 seconds. 2. Circulate during pair work. At each pair, ask ONE probing question — rotate through: 'What does the 0.25 represent in terms of the pizza?' / 'Why are you multiplying by r TWICE?' / 'For the segment, what shape are you subtracting and why?' / 'Does the simulation	Strategy: THREE-SOURCE TRIANGULATION. Learners compare three independent sources of the same answer (hand calculation, calculator, simulation). This mirrors scientific practice (NGSS SEP 4: Analysing and Interpreting Data) — a single source can be wrong, but agreement across three sources builds justified confidence. Connecting to the phenomenon: Say: 'When the Nairobi pizza chef cuts a chapati into slices and charges by size, he needs to be certain his area calculation is right — one method is not enough. We	Observable indicator: The three-column comparison table is the artefact. Teacher checks that (a) every intermediate calculator step is recorded (not just the final answer), (b) the segment column shows a subtraction step, and (c) the 'reason for mismatch' column contains a mathematical explanation, not just 'I made a mistake'. Flag pairs whose simulation result matches the calculator but both differ from their hand calculation — these pairs need targeted support in Phase 3.

for similar readings	offline GeoGebra applet (or printed dynamic geometry screenshot), learners input the same radius and angle values and read off the computed area. Step 3 — Three-column comparison table: Learners record (i) their original hand-calculation from prior lessons, (ii) their new calculator result, (iii) the simulation result, and write a 'Match / Mismatch' verdict with a reason for any discrepancy.	agree? If not, which do you trust — and why?' 3. At the 25-minute mark, STOP the class. Say: 'Hands up — which pair found a mismatch between your old hand-calculation and the calculator today?' Count hands publicly. Say: 'Good — mismatches are data, not failures. We will diagnose them next.' 4. Ask one pair with a mismatch to project/read their three-column table. Ask the class: 'At which step did the error enter?' WAIT TIME: 12 seconds. Cold-call 3 learners for hypotheses before revealing.	are doing what a careful engineer or farmer would do: checking with multiple tools.'	
Explain Phase 📹 **VIDEO:** Radians as ratio of arc length to radius Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Quadratic Formula (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Whole-class Error Autopsy. The teacher presents four 'Patient Solutions' (deliberately flawed worked examples on the board/ARES screen). For each patient, learner pairs must: (1) DIAGNOSE — identify the exact line where the error occurs; (2) EXPLAIN — write one sentence naming the error type; (3) PRESCRIBE — rewrite the correct solution from the error line onward. Error types embedded: Patient 1 (Sector): Learner used θ in place of θ/360 — i.e., wrote $A = 120 \times \pi \times 7^2$ instead of $A = (120/360) \times \pi \times 7^2$. Patient 2 (Sector): Learner used diameter (28 cm) instead of radius (14 cm) in $A = (\theta/360) \times \pi \times 28^2$. Patient 3 (Segment): Learner correctly found sector area but forgot to subtract the triangle: final answer = sector area only. Patient 4 (Segment): Learner rounded π to 3 (not 3.14159) at the first step, causing a large cumulative error. After each 'Patient', one pair presents their diagnosis. Class votes: 'Agree / Disagree / Modify' by show of hands. Real-world anchoring: Each patient is framed	1. Distribute the 'Error Autopsy' sheet. Say: 'You are mathematicians acting as doctors. Each patient solution is sick. Your job: find the disease, name it, and cure it. Work in your pairs. You have 12 minutes for all four patients.' 2. After 12 minutes, cold-call one pair per patient to present. For each presentation: (a) ask the class 'Thumbs up if you agree with the diagnosis — thumbs sideways if you want to modify it.' WAIT TIME: 10 seconds before counting thumbs. (b) Ask a second pair: 'Can you add anything to the prescription — the corrected working?' 3. For Patient 3 (missing triangle subtraction), ask: 'Connect this back to our chapati. If you gave someone the sector area and called it the segment area, would they be getting more or less chapati than they paid for? By how much?' Cold-call 2 learners. WAIT TIME: 12 seconds. 4. After all four patients, summarise on the board: write the four error names as a class-generated 'Most Wanted Errors' list. Say: 'These are the four errors that will cost marks —	Strategy: ERROR AUTOPSY (Structured Misconception Analysis). Rather than re-teaching correct procedures, this strategy places learners in the role of expert evaluators — activating deeper processing (NGSS SEP 6: Constructing Explanations). Research shows learners who explain why an answer is wrong retain the correct procedure more durably than those who only practise correct examples. Connecting to the phenomenon: The teacher says: 'Every one of these four errors could have been made by someone trying to calculate the area of that first unequal pizza slice we saw on Day 1. Which error would have given the most wildly wrong answer for the wide slice? Let's estimate.' This keeps the pizza phenomenon alive as a reference anchor.	Observable indicator: The 'Diagnose / Explain / Prescribe' written responses on the Error Autopsy sheet. Teacher looks for: (a) the error is located at a specific line/step, not described vaguely; (b) the explanation names the mathematical concept involved (e.g., 'confused diameter with radius', 'omitted the formula for area of a triangle'); (c) the corrected working is complete and reaches the right numerical answer. Any pair that cannot diagnose Patient 3 (segment = sector only) receives an immediate targeted intervention: teacher draws a sector and a triangle on the pair's desk and asks 'What is shaded here that should NOT be in the segment?'

	in a Kenyan context — Patient 1: a Kericho tea farmer calculating an irrigation sector; Patient 2: a Nairobi architect designing a circular stage; Patient 3 & 4: a Kisumu fisherman calculating a net segment.	and in real life, cost a farmer or architect real money. Memorise the names, not just the fixes.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Radians as ratio of arc length to radius] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Quadratic Formula (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board, which has accumulated questions and partial answers from Lessons 1–8. Each learner individually writes two DQB contributions on separate cards: (1) an ANSWER card — one confirmed thing they can now state with certainty about calculating sector or segment area (supported by their digital verification); (2) a REMAINING QUESTION card — one thing still uncertain or that they want to explore in Lesson 10 (e.g., 'Can this formula work for the curved part of a Maasai boma fence layout?', 'What happens when θ = 180° — is the segment a semicircle?'). Learners pin their cards on the DQB under the headings 'We Now Know' and 'We Still Wonder'. The class reviews the 'We Now Know' column and the teacher facilitates a 3-minute gallery walk — learners read each other's answer cards and place a tick sticker on any card they agree with and can also justify.	1. Direct learners to the DQB. Say: 'Look at the questions we posted on Day 1. Pick one question that you can now fully answer — write the answer on a green card with your evidence. Then write one question we have NOT yet fully answered on a yellow card.' Allow 5 minutes for writing. 2. During pinning, circulate and read cards. Identify 2–3 Answer cards that are precise and well-evidenced — ask those learners to read theirs aloud to the class. 3. For the gallery walk, say: 'Move around. Put a tick on any green card whose answer you could defend to a classmate. If you disagree with a card, put a question mark — but be ready to say why.' Allow 3 minutes. 4. After the gallery walk, point to 2 question-marked cards. Cold-call the learner who placed the question mark: 'Tell us your concern in one sentence.' WAIT TIME: 10 seconds. Then ask the card's author to respond. 5. Close by pointing at the Driving Question banner and saying: 'We started this unit because a pizza was cut into unequal slices and we could not explain why bigger angles meant bigger areas by an exact amount. Look at this DQB — what CAN we explain now that we could not on Day 1?'	Strategy: DRIVING QUESTION BOARD REVISION (Collaborative Knowledge Tracking). The DQB is a living document that makes the class's collective epistemic progress visible. Updating it at Lesson 9 (of 10) allows learners to see how far they have come — converting 'wonders' into 'knowns' — while surfacing any residual gaps before the summative Lesson 10. This mirrors NGSS SEP 1 (Asking Questions) and SEP 7 (Engaging in Argument from Evidence). Connecting to the phenomenon: The teacher explicitly connects the 'We Now Know' column to the original pizza question: 'Our first wonder was — is a double-angle slice exactly double the area? We can now answer that with a formula AND a calculator check.'	Observable indicator: Quality of the green Answer cards. Teacher checks that answers are: (a) stated as a general rule, not just about one specific calculation; (b) include a formula or worked numeric example as evidence; (c) correctly distinguish between sector and segment where relevant. Tick-sticker counts give the teacher a quick class-consensus heatmap. Cards with zero ticks are flagged for whole-class discussion in Lesson 10's opening review.
Model Building Phase	Learners individually produce a final Annotated Formula Map — a single A4 diagram that shows: (1)	1. Say: 'Your Formula Map is your personal mathematical tool — like a calibrated measuring instrument.	Strategy: ANNOTATED MODEL CONSOLIDATION (Personal Reference Artefact). The Formula	Observable indicator: The submitted Formula Map is assessed against five checkpoints:

📹 **VIDEO:** [Radians as ratio of arc length to radius](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Quadratic Formula (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	a labelled circle with a sector and a segment clearly marked; (2) the sector area formula A = (θ/360°) × πr² with each symbol annotated (θ = central angle in DEGREES, r = radius, π ≈ 3.14159); (3) the segment area formula: A_segment = A_sector − A_triangle, with the triangle area formula ½r²sinθ also written; (4) a 'Common Errors' warning box listing the four errors from the Error Autopsy; (5) one Kenyan real-world example of each (sector: chapati slice, r = 21 cm, θ = 90°; segment: irrigation arc, r = 35 m, θ = 60°) with the final numerical answer confirmed by calculator. Learners submit this map as an exit ticket. Final 2 minutes: teacher asks the class to look at the pizza phenomenon image (displayed on screen) and cold-calls 3 learners: 'Using everything on your Formula Map — can you now predict the exact area of ANY slice from that pizza if I give you the angle? Yes or no, and justify.'	It must be precise enough that a classmate who missed this unit could use it to solve any sector or segment problem.' Allow 7 minutes for drawing and annotation. Circulate and check that EVERY map includes the 'degrees' annotation on θ and the segment subtraction step — these are the two most commonly missing elements. 2. For any learner who finishes early, say: 'Add a third real-world Kenyan example — not chapati or pizza. Think: a Maasai boma fence arc, a Lake Victoria fishing net sector, a Kisumu stadium seating arc.' 3. With 3 minutes remaining, collect maps as exit tickets. Display the original phenomenon image. Cold-call 3 learners (spread across ability groups): 'If this pizza has radius 28 cm and this wide slice has a central angle of 135°, use your Formula Map — what is the area of that slice to 2 decimal places?' WAIT TIME: 15 seconds per learner. 4. Confirm correct answer (A = (135/360) × π × 28² ≈ 923.63 cm²). Say: 'On Day 1 we could not answer that question. Today you can — with confidence and with proof from two independent tools.'	Map serves as both a summative consolidation artefact and a self-regulation tool — learners must decide what is essential enough to include, forcing prioritisation of key ideas (NGSS SEP 2: Developing and Using Models). The act of annotating symbols (especially 'θ in DEGREES') directly addresses the most persistent error identified in Phase 3. Connecting to the phenomenon: The final cold-call returns explicitly to the original pizza image, completing the narrative arc: from 'we could not explain it' (Lesson 1) to 'we can calculate any slice with precision and verify it digitally' (Lesson 9).	(1) sector formula present and correctly written; (2) segment formula present with subtraction step shown; (3) θ annotated as 'degrees'; (4) at least one Kenyan numerical example fully worked; (5) 'Common Errors' warning box included. Teacher sorts maps into three piles after class: Complete (all 5 checkpoints), Partial (3–4 checkpoints), Incomplete (0–2 checkpoints). Partial and Incomplete learners receive targeted written feedback before Lesson 10. The cold-call final calculation also gives a live indicator of fluency — inability to start the calculation identifies learners who need a pre-Lesson-10 mini-intervention.

D. TEACHER REFLECTION

After Lesson 9, reflect on the following: (1) ERROR PATTERNS — Which of the four 'Most Wanted Errors' appeared most frequently in the Error Autopsy responses and in the Formula Maps? Did learners correctly identify the degrees-vs-radians confusion, or did many still write the formula without annotating that θ must be in degrees? (2) DIGITAL TOOL CONFIDENCE — Were learners able to use the scientific calculator fluently to compute (θ/360) × π × r² in a single sequence, or did many reach incorrect intermediate values? If calculator use was laboured, consider a 5-minute calculator drill at the start of Lesson 10. (3) DQB QUALITY — Were the 'We Now Know' cards genuinely generalised (e.g., 'The area of any sector is proportional to its central angle') or still example-specific (e.g., 'The chapati slice with 90° has area 346.5 cm²')? If the latter, plan a brief generalisation prompt at the start of Lesson 10. (4) FORMULA MAP COMPLETENESS — How many learners were in the 'Partial' or 'Incomplete' pile? These learners must receive written feedback and, if possible, a brief face-to-face check before Lesson 10's summative application. (5) PHENOMENON CONNECTION — Did learners spontaneously reference the pizza/chapati phenomenon during the Error Autopsy, or only when prompted? Strong spontaneous connections suggest deep anchoring; absence suggests the teacher should explicitly re-invoke the phenomenon at the start of Lesson 10.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed that when the same sector and segment calculations from earlier lessons were re-run on scientific calculators and checked against a digital simulation, results either confirmed their hand calculations or revealed specific, diagnosable errors. They also observed four annotated 'Patient Solutions' on the board — flawed worked examples drawn from Kenyan real-world contexts (a Kericho irrigation sector, a Nairobi architectural arc, a Kisumu fishing-net segment) — and saw exactly which line of working introduced each error.
What did I learn?	Students learned that: (1) the sector area formula $A = (\theta/360°) \times \pi r^2$ requires θ to be measured in degrees, not radians, and that confusing the two produces a wildly incorrect answer; (2) the segment area requires an explicit subtraction of the triangle area ($A_{segment} = A_{sector} - \frac{1}{2}r^2\sin\theta$), and omitting this step gives the sector area, not the segment; (3) using diameter instead of radius in the formula inflates the answer by a factor of 4; (4) premature rounding of π introduces cumulative error in the final answer; (5) digital tools (calculators, simulations) provide independent verification but do not replace the need to understand each step — a calculator will faithfully compute the wrong formula if the wrong formula is entered.
How does this explain the phenomenon?	Students explained their understanding by: (1) diagnosing and correcting four flawed worked examples in the Error Autopsy activity, naming the specific error type at the exact line it occurred; (2) producing an Annotated Formula Map linking the sector and segment formulas to labelled diagrams, Kenyan numerical examples, and a 'Common Errors' warning box; (3) updating the Driving Question Board with 'We Now Know' cards that stated generalised rules (not just specific examples) and justifying their answers with formula-based evidence; (4) answering the final cold-call calculation — finding the area of a pizza slice with $r = 28$ cm and $\theta = 135°$ to two decimal places — demonstrating confident, verified application of the formula to the original anchoring phenomenon.

LESSON 10 (40 minutes): Assessment and Reflection: Connecting the Pizza Phenomenon to All Learned Concepts	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate and assess learners ability to calculate arc length, sector area, and segment area using the radius and central angle, and to reflect on how the anchoring pizza phenomenon underpins all learned concepts.
Knowledge	Learners will demonstrate knowledge of the formulas: arc length = (theta/360) times 2 pi r, sector area = (theta/360) times pi r squared, and segment area = sector area minus half r squared sin theta; they will distinguish between minor and major sectors and segments and recall common errors to avoid.
Skills	Learners will apply all three formulas in a multi-step summative task involving a circular shamba, present their solutions clearly with correct units, and reflect critically on how each formula connects back to the original pizza-slicing phenomenon.
Attitudes	Learners will demonstrate confidence in mathematical reasoning, perseverance when solving multi-step problems, and appreciation of how mathematics describes real-world situations in Kenyan farming and food contexts.
Key Inquiry Question	How do we calculate the area of a part of a circle, and how do areas of sectors and segments of a circle apply in day-to-day activities in Kenya?
Purpose in Storyline	This final lesson closes the inquiry storyline by requiring learners to independently demonstrate mastery of every formula derived across Lessons 3 to 9, explicitly tracing each formula back to the pizza-slicing phenomenon and updating the Driving Question Board with final evidence-based answers to all posted questions.
Safety Notes	No physical hazards are anticipated. If a physical pizza or chapati is used for the closing reflection, ensure food handling hygiene. Remind learners to handle rulers, protractors, and calculators with care.

B. LESSON OVERVIEW
Lesson 10 is the summative and reflective capstone of the sub-strand. The teacher briefly re-presents the original anchoring phenomenon (a circular pizza cut into unequal slices) and then releases learners to complete a structured individual summative task in which they are given the radius and central angle of a sector of a circular shamba and must calculate arc length, sector area (minor), major sector area, minor segment area, and major segment area, showing all working. Following the task, selected learners present their solutions, the class identifies and discusses errors, and the Driving Question Board is finalised with evidence-based answers. The lesson ends with a structured whole-class reflection that explicitly maps the pizza-slicing experience onto every formula learned.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: Area Model Algebra Source: PhET	Learners re-encounter the original phenomenon. The teacher places (or projects an image of) a circular pizza or round chapati on the table and reminds learners of the very	Teacher re-displays or physically holds up the circular pizza and says: Cast your mind back to Lesson 1. We stood around this pizza and asked: is a wider-angled	Anchored prediction using the original phenomenon to activate prior knowledge; partner discussion to surface misconceptions before the	Teacher reads 4 to 5 exercise books during circulation to note whether learners correctly reason that 80 degrees is less than 180 degrees so the sector is the minor

Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Length Measurements to a Fraction of an Inch (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	first lesson. A large circular diagram is drawn on the board representing a circular shamba in Nakuru with radius 21 m and a fenced sector of central angle 80 degrees. Learners are asked: Before we calculate, predict whether the fenced sector or the unfenced sector is larger. Also predict: will the segment cut off by the chord of the fenced sector be more than half or less than half of the sector? Learners write their predictions in their exercise books with a brief reason, then share with a partner for 1 minute.	slice always exactly proportionally larger? Today you will prove your answer using every tool we have built together. Then the teacher draws the shamba diagram on the board, labels radius as 21 m and central angle as 80 degrees, and asks: Write your prediction now. Is the fenced sector more than or less than half the shamba? Is the segment more than or less than the sector? You have 2 minutes. WAIT TIME: 2 minutes of silent individual writing. Teacher circulates, reading predictions without commenting. After sharing, teacher says: Hold your predictions. By the end of this lesson, your calculations will confirm or correct them.	summative task begins; prediction-recording ensures accountability.	sector (less than half). Learners who say it is more than half are flagged for closer monitoring during the task.
Observe Phase 📹 **VIDEO:** Video: Area Model Algebra Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Length Measurements to a Fraction of an Inch (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive the printed or board-displayed summative task sheet. They read through all five parts silently for 2 minutes, identifying what is given and what is required for each part. They then annotate the shamba diagram by labelling the radius, central angle, chord, minor sector, major sector, minor segment, and major segment. This annotation phase mirrors Lesson 2 in which they labelled parts of circular cut-outs. Learners also recall from Lesson 9 the common errors list (using diameter instead of radius, confusing degrees and radians, omitting the triangle subtraction, premature rounding) and write those four error warnings at the top of their working page as a self-checklist.	Teacher displays the task and says: Before you write a single number, read every question carefully and annotate the diagram. Label everything you can see. You have 2 minutes of silent reading. WAIT TIME: 2 minutes. Teacher then says: Now write the four common errors we identified in Lesson 9 at the top of your page. These are your guardrails. Use pi as 22/7 or 3.14159 consistently throughout, whichever you choose, but do not switch. Then teacher asks one learner to read the task aloud so all learners hear it once more, and clarifies any vocabulary (for example, what fenced sector means in the shamba context).	Diagram annotation as an observation strategy mirrors the opening lesson and reinforces the vocabulary consolidation from Lesson 2; self-checklist of common errors activates metacognitive monitoring from Lesson 9.	Teacher checks that every learner has labelled minor sector, major sector, minor segment, and major segment correctly on the diagram before they begin computing. Learners who mislabel are redirected with the question: Which sector has an angle less than 180 degrees?
Explain Phase 📹 **VIDEO:** Video: Area Model Algebra	Learners independently complete the five-part summative task over approximately 18 minutes. The five parts are: Part A — calculate the	Teacher says: You have 18 minutes. Work independently and silently. Show every step of working — a correct answer	Independent summative task followed by peer critique of presented solutions; explicit connection of each formula to the	Teacher collects all exercise books at the end of the lesson for marking against a provided rubric. During presentations, teacher uses

Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Length Measurements to a Fraction of an Inch (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	arc length of the minor sector (answer: 29.33 m using pi = 22/7); Part B — calculate the area of the minor sector (answer: 307.9 square metres); Part C — calculate the area of the major sector (answer: 1077.6 square metres, cross-checked as total circle area minus minor sector area); Part D — calculate the area of the minor segment (answer: minor sector area minus half times 21 squared times sin 80 degrees, giving approximately 307.9 minus 216.5 = 91.4 square metres); Part E — reflect: a farmer wants to plant maize on the minor segment only. Is this area enough to plant 200 maize plants if each plant needs 0.4 square metres? (Answer: 91.4 divided by 0.4 = 228.5, so yes, there is enough space.) After individual work, four learners are selected to present one part each on the board, showing full working. The class listens, asks questions, and identifies any errors using the four-error checklist.	without working earns partial marks only. Refer to your error checklist constantly. WAIT TIME: 18 minutes of monitored independent work. Teacher circulates continuously, noting specific errors without correcting them (to preserve summative integrity) but recording names for targeted follow-up. At the 10-minute mark teacher gives a quiet time check: You should be finishing Part C or starting Part D. After the working period, teacher selects four presenters and says: As each person presents, your job is to use the four-error checklist to spot any mistakes. Raise your hand silently if you find one. After each presentation teacher asks the class: Did the presenter avoid all four errors? Can someone explain why sin 80 degrees is used in Part D and not sin 40 degrees? WAIT TIME: 30 seconds after each explanation prompt.	pizza-slicing phenomenon through the teacher narrating after each part: In Lesson 1 you estimated this; now you can calculate it exactly; error-checklist-guided peer review reinforces metacognitive consolidation from Lesson 9.	cold-calling to ask non-presenting learners to verify specific calculation steps, ensuring broad engagement. Common errors observed during circulation are noted and used to populate the lesson reflection notes.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Area Model Algebra Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Length Measurements to a Fraction of an Inch (Flexbook)	With 7 minutes remaining, the class gathers around the Driving Question Board which has been displayed since Lesson 1. Learners re-read all questions posted across the unit. The teacher guides the class to post final answer cards next to each question, using evidence from their summative task working. For example: the question Is a slice with twice the angle always exactly twice the area? receives the answer card Yes — because both arc length and sector area are directly proportional to the central angle, as shown by the formula (theta/360) times pi r squared. The	Teacher stands at the DQB and says: Every question we posted in Lesson 1 deserves a final evidence-based answer today. Let us go through them one by one. For each question, teacher asks: Who can read the question? Who has evidence from their task today that answers this? Can you connect that evidence back to the pizza we started with? WAIT TIME: 20 seconds after each question before accepting responses. Teacher posts answer cards collaboratively with learner input, narrating: In Lesson 1 we could only guess. In Lesson 10 we can prove. That is the journey of this	Collaborative DQB finalisation as a public record of knowledge growth; explicit tracing from anchoring phenomenon (pizza) through each lesson milestone to the final summative task creates narrative coherence across the sub-strand; personal reflection cards surface individual metacognition.	Teacher scans personal reflection cards to assess depth of metacognitive engagement. Learners who write only formulaic responses (for example, I learned the formula) are identified for follow-up questioning: Can you tell me why the formula works and not just what it is?

Source: CK-12 🔍 Search ARES for similar readings	question Can we predict the exact area of any slice? receives the answer card Yes — using A = (theta/360) times pi r squared, confirmed by our shamba calculation today. Learners also post a Kenya-context card: The same formula used for the pizza in Lesson 1 calculated the maize-planting area of a circular shamba in Nakuru in Lesson 10. Learners take 2 minutes to write one personal DQB reflection card: the most surprising thing they learned during the unit.	entire unit. Teacher ensures the final DQB visually shows a complete arc from phenomenon to formula to real-life application.		
Model Building Phase 📹 **VIDEO:** Video: Area Model Algebra Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Length Measurements to a Fraction of an Inch (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final 5 minutes, learners update their individual concept maps (started in Lesson 4) to include all formulas, their relationships, and at least two Kenyan real-life applications per formula. They draw arrows connecting: pizza slice to sector area formula to circular shamba to maize planting. They add a final reflection sentence at the bottom of the map: The driving question is answered because we can use radius and central angle to calculate the exact area of any circular slice using A = (theta/360) times pi r squared, and we can extend this to segments by subtracting half r squared sin theta. Learners share their updated concept maps with a neighbour for a 60-second gallery-walk-style peer review, then the teacher closes the lesson with a whole-class verbal celebration of the learning journey from Lesson 1 to Lesson 10.	Teacher says: Open your concept maps from Lesson 4. You have 3 minutes to add everything we have learned since then — all formulas, all Kenyan contexts, all connections to the pizza. WAIT TIME: 3 minutes silent individual work. Teacher then says: Share your map with the person next to you. Tell them in 60 seconds: what is the most powerful connection on your map? WAIT TIME: 60 seconds. Teacher closes by saying: In Lesson 1, someone asked: can we predict the exact area of any pizza slice if we only know the radius and angle? Look at what you have done today. You calculated the arc length, the sector area, the segment area, and told a farmer in Nakuru whether he has enough land for 200 maize plants. You answered the driving question. That is mathematics in action in Kenya.	Concept map update as a model-building artifact that visually consolidates the entire sub-strand; peer sharing activates verbal explanation as a consolidation strategy; teacher closing narrative explicitly answers the driving question using the summative task evidence, completing the storyline from Lesson 1.	Teacher quickly reviews 5 to 6 concept maps during the peer-share to check that learners have included correct formulas, at least one Kenya context, and the pizza-to-shamba connection. Maps that are incomplete or contain formula errors are noted for follow-up feedback during book return in the next session.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners independently apply all three formulas without scaffolding, or did many revert to formula confusion under summative conditions? (2) Were the four common errors from Lesson 9 successfully internalised — did learners self-correct using their checklists? (3) Did the DQB finalisation reveal any questions that remain unanswered or partially answered, signalling gaps for future teaching? (4) Did learners make explicit and unprompted connections between the pizza-slicing phenomenon and the shamba calculation, or did those connections require heavy teacher prompting? (5) Were Kenyan contexts (Nakuru shamba, maize planting, circular irrigation) sufficient to make the mathematics feel relevant and purposeful to all learners? (6) Which learners struggled most with Part D (segment area), and what reteaching support will you provide before the next sub-strand begins? Note specific learner names, their errors, and planned interventions in your professional journal.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed a circular pizza cut into unequal slices in Lesson 1 and noticed that wider-angled slices appeared larger. Today we observed a circular shamba in Nakuru divided into a fenced sector of 80 degrees and an unfenced major sector, and we measured (calculated) the arc length, sector areas, and segment areas using radius 21 m and central angle 80 degrees.
What did I learn?	We learned that arc length = (theta/360) times 2 pi r; sector area = (theta/360) times pi r squared; major sector area = total circle area minus minor sector area; segment area = sector area minus half r squared sin theta; and major segment area = total circle area minus minor segment area. We also learned that these formulas apply directly to real Kenyan contexts including pizza pricing, shamba allocation, maize planting density, and circular irrigation field management.
How does this explain the phenomenon?	We explained why a pizza slice with twice the angle is exactly twice the area because both quantities are directly proportional to the central angle as a fraction of 360 degrees. We explained why the segment formula requires subtracting the triangle area and how omitting that step gives the sector area by mistake. We explained how every formula we derived traces back to the original question posed in Lesson 1: can we predict the exact area of any slice if we know only the radius and the angle of the cut?

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.